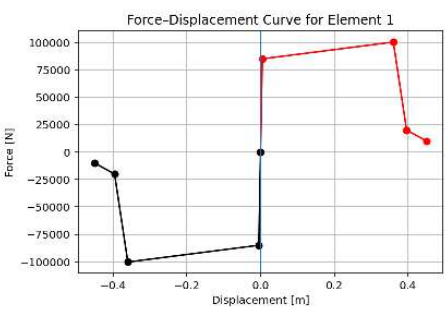
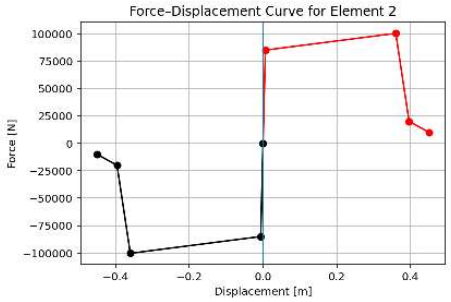
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM SOIL STRUCTURE SYSTEM: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

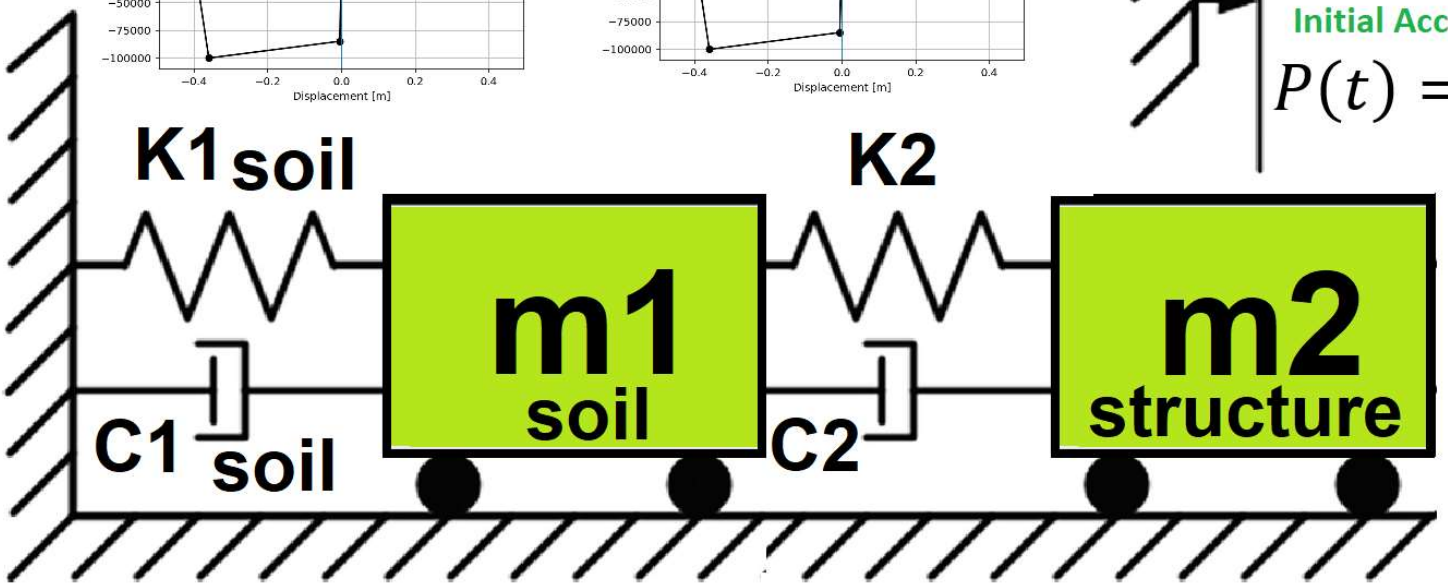


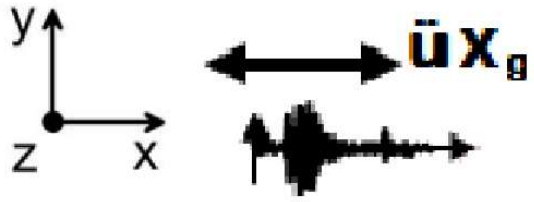


INCREMENTAL DISPLACEMENT

Initial Displacement
Initial Velocity
Initial Acceleration

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$





$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

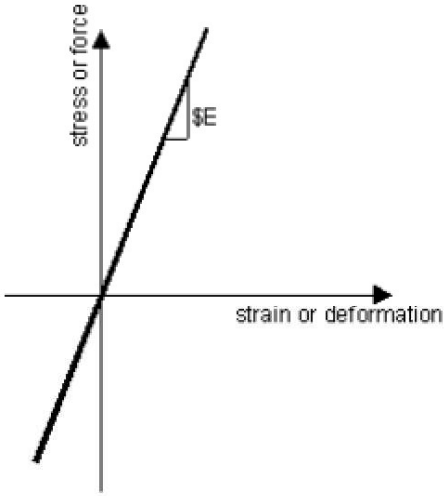
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

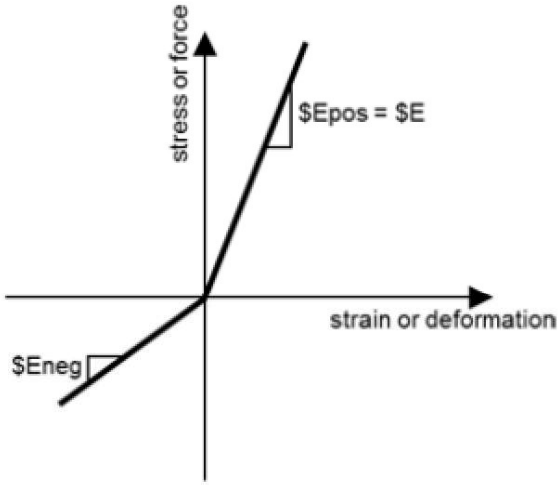
$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

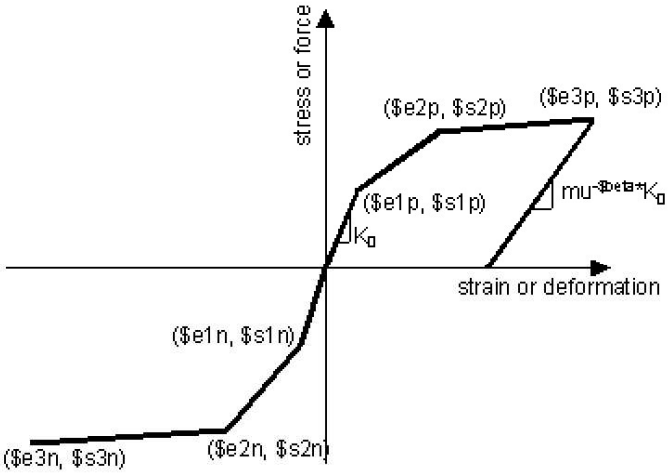
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



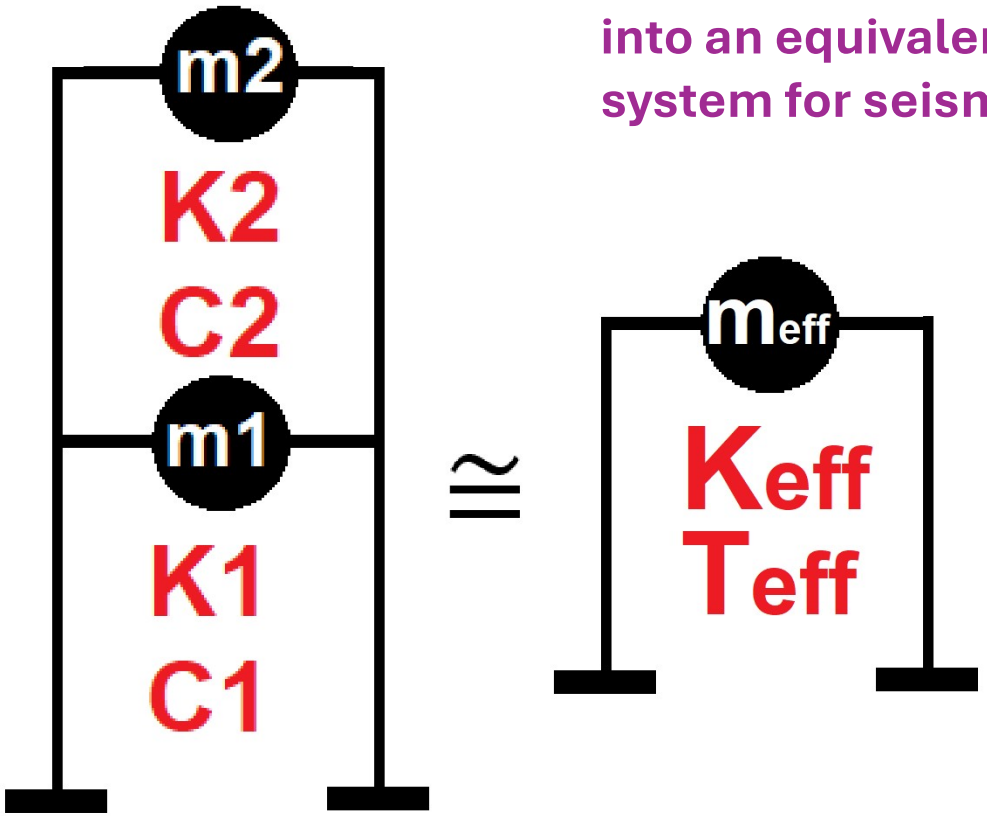


ELASTIC MATERIAL



INELASTIC MATERIAL

Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

Spyder (Python 3.12)

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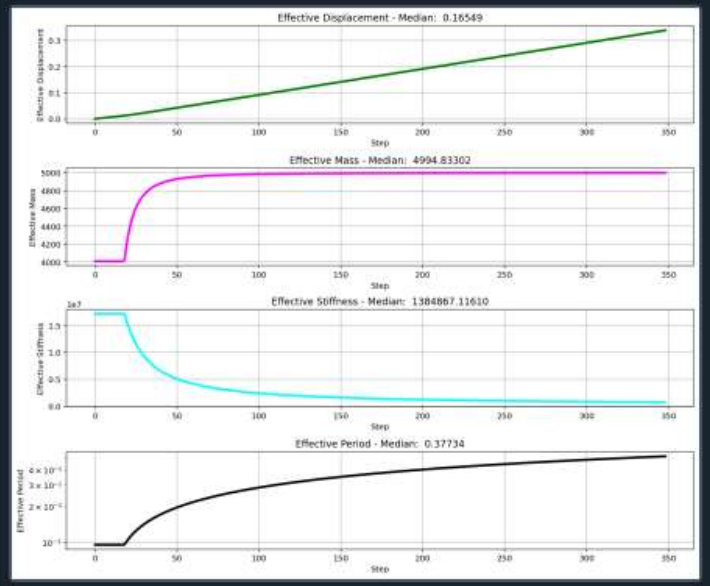
...GE_INDEX_&_ENERGY_DISSIPATION_CAPACITY_INDEX\22_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TURE_MDOF_8_ANA\P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

```
1  ##########################################################################################
2  #                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3  # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM SOIL STRUCTURE SYST
4  # FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-
5  #----------------------------------------------------------------------------------------
6  #                  EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANAL
7  #----------------------------------------------------------------------------------------
8  #                                          P(t) = P0 sin(wt)
9  #                                      P(t) = P0 exp(-0.05wt) sin(wt)
10 #----------------------------------------------------------------------------------------
11 #         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
12 #                                     OF ENERGY DISSIPATION CAPACITY INDEX
13 #----------------------------------------------------------------------------------------
14 #                          THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
15 #                                  EMAIL: salar.d.ghashghaei@gmail.com
16 ##########################################################################################
17 """
18 Nonlinear Seismic Performance Assessment of a MDOF Soil Structure System:
19 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
20
21 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF S
22  for performance-based earthquake engineering. The 2D model incorporates both material non
23  (elastic-perfectly plastic or hysteretic steel with strain hardening)
24  and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
25  and large displacements—essential for collapse assessment.
26
27 Eight analysis protocols are implemented:
28 (1) [PERIOD] : Structural Period
29 (2) [STATIC] : Gravity load analysis establishing dead load state
30 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
31  and plastic mechanism identification
32 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
33  pinching behavior, and energy dissipation degradation
34 (5) [STATIC EXTERNAL TIME-DEPENDENT LOADING] : Static Analysis of External time-dependent
```



26 %

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Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 9, Col 39 UTF-8 CRLF RW Mem 54%

STATIC ANALYSIS

Spyder (Python 3.12)

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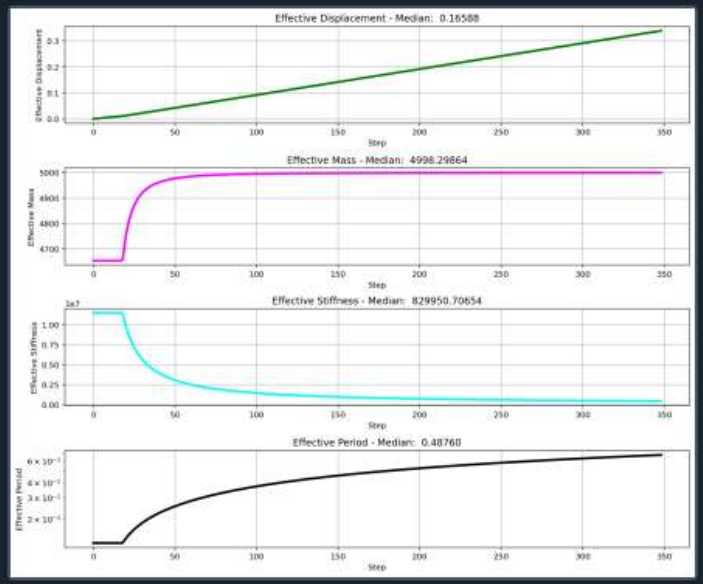
P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

```
221     if ANAL_TYPE == 'STATIC':
222         ops.integrator('LoadControl', 1.0)
223         ops.analysis('Static')
224         OK = ops.analyze(1)
225         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
226         ops.reactions()
227         reaction.append(ops.nodeReaction(1, 1))                    # BASE REACTION
228         disp.append(ops.nodeDisp(center_node, 1))                  # DISPLACEMENT NOD
229         DI.append(100*(np.abs(disp[-1])-DY)/(DSU-DY))              # DAMAGE INDEX
230         if DI[-1] <= 0.0: # DAMAGE INDEX -> 0.0 <= DI <=100
231             DI[-1] = 0.0
232         if DI[-1] >= 100:
233             DI[-1] = 100.0
234         print('\n\nSTATIC ANALYSIS DONE.\n\n')
235
236         DATA = (reaction, disp, DI)
237
238         return  DATA
239
240     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
241         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
242         DINCR = -0.001  # [m] Incremental Vertical Displacement
243         DMAX = -0.35    # [m] Max. Displacement
244         ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
245         ops.analysis('Static')
246         Nsteps =  int(np.abs(DMAX/ DINCR))
247         STEP = 0.0
248         for step in range(Nsteps):
249             OK = ops.analyze(1)
250             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
251             ops.reactions()
252             reaction.append(ops.nodeReaction(1, 1))                # BASE REACTIO
253             disp.append(ops.nodeDisp(center_node, 1))              # DISPLACEMENT
254             # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
```



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PUSHOVER ANALYSIS

Spyder (Python 3.12)

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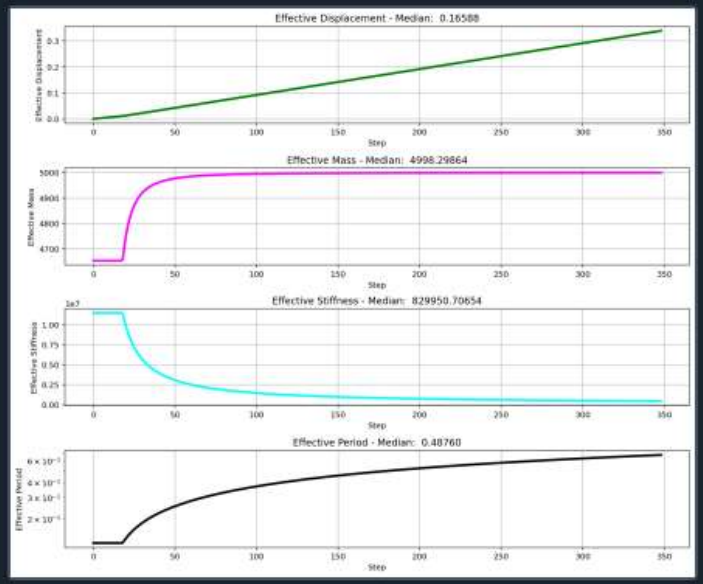
P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

```
221     if ANAL_TYPE == 'STATIC':
222         ops.integrator('LoadControl', 1.0)
223         ops.analysis('Static')
224         OK = ops.analyze(1)
225         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
226         ops.reactions()
227         reaction.append(ops.nodeReaction(1, 1))                    # BASE REACTION
228         disp.append(ops.nodeDisp(center_node, 1))                  # DISPLACEMENT NOD
229         DI.append(100*(np.abs(disp[-1])-DY)/(DSU-DY))              # DAMAGE INDEX
230         if DI[-1] <= 0.0: # DAMAGE INDEX -> 0.0 <= DI <=100
231             DI[-1] = 0.0
232         if DI[-1] >= 100:
233             DI[-1] = 100.0
234         print('\n\nSTATIC ANALYSIS DONE.\n\n')
235
236         DATA = (reaction, disp, DI)
237
238         return  DATA
239
240     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
241         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
242         DINCR = -0.001  # [m] Incremental Vertical Displacement
243         DMAX = -0.35    # [m] Max. Displacement
244         ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
245         ops.analysis('Static')
246         Nsteps =  int(np.abs(DMAX/ DINCR))
247         STEP = 0.0
248         for step in range(Nsteps):
249             OK = ops.analyze(1)
250             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
251             ops.reactions()
252             reaction.append(ops.nodeReaction(1, 1))                # BASE REACTIO
253             disp.append(ops.nodeDisp(center_node, 1))              # DISPLACEMENT
254             # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
```



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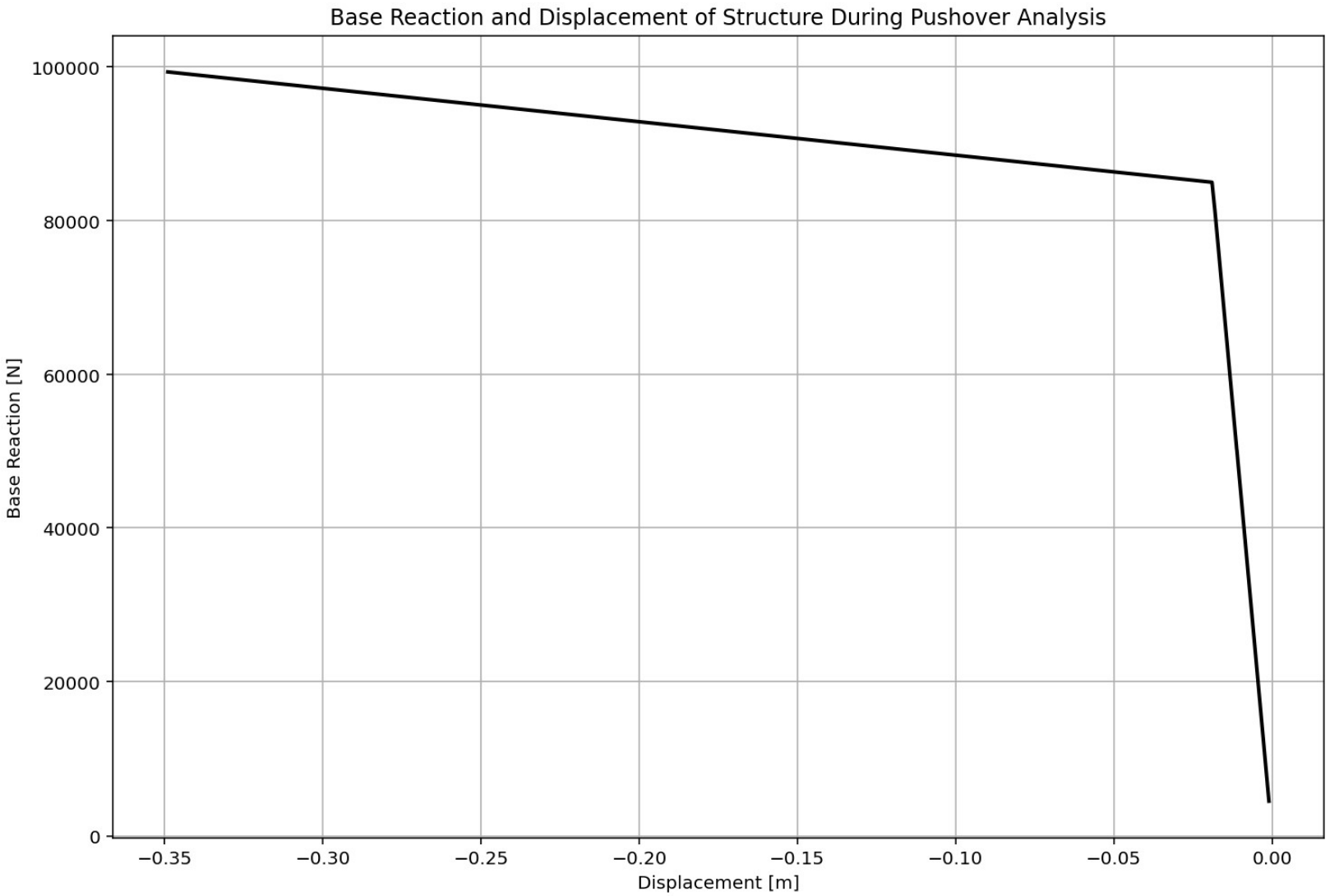
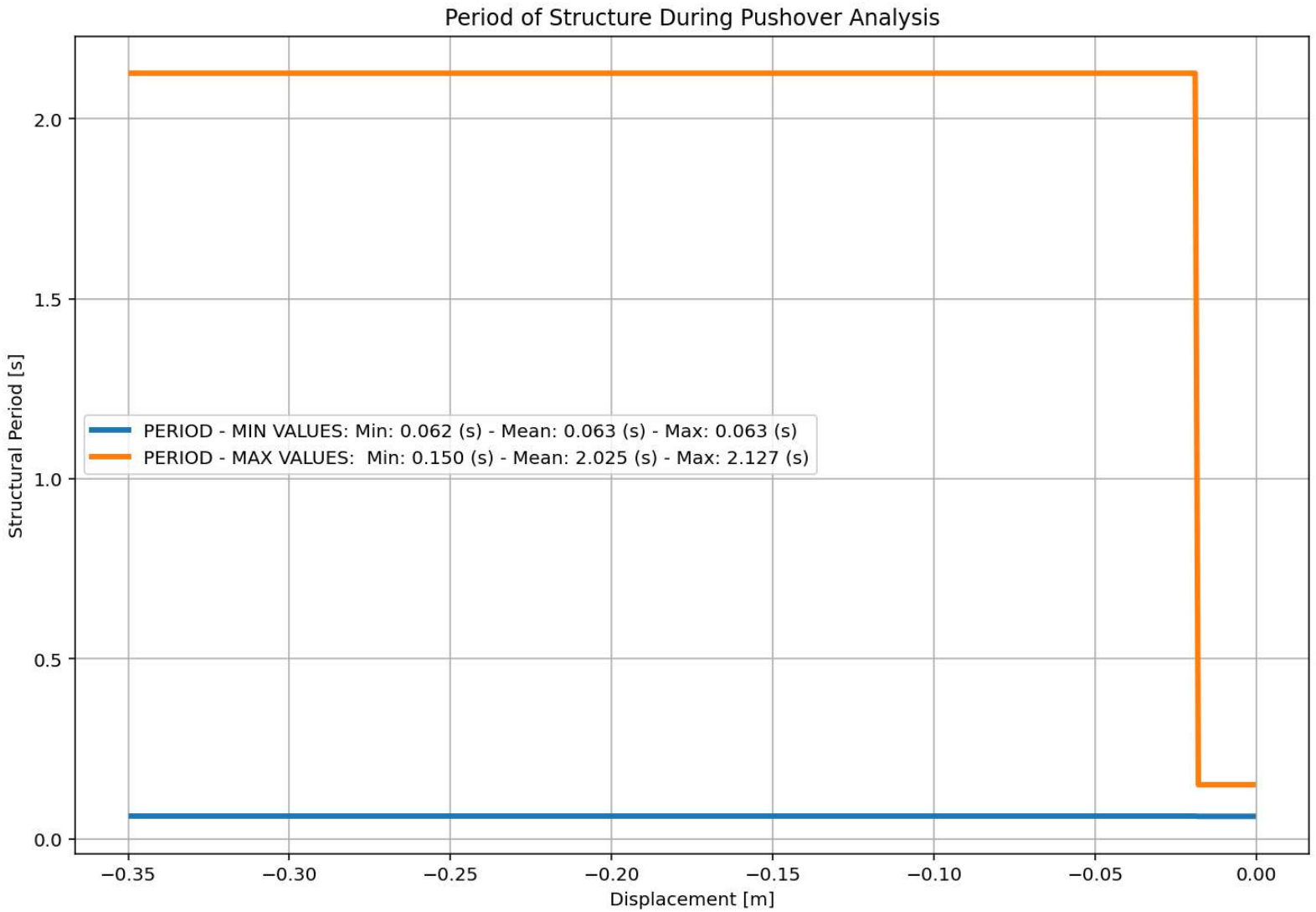
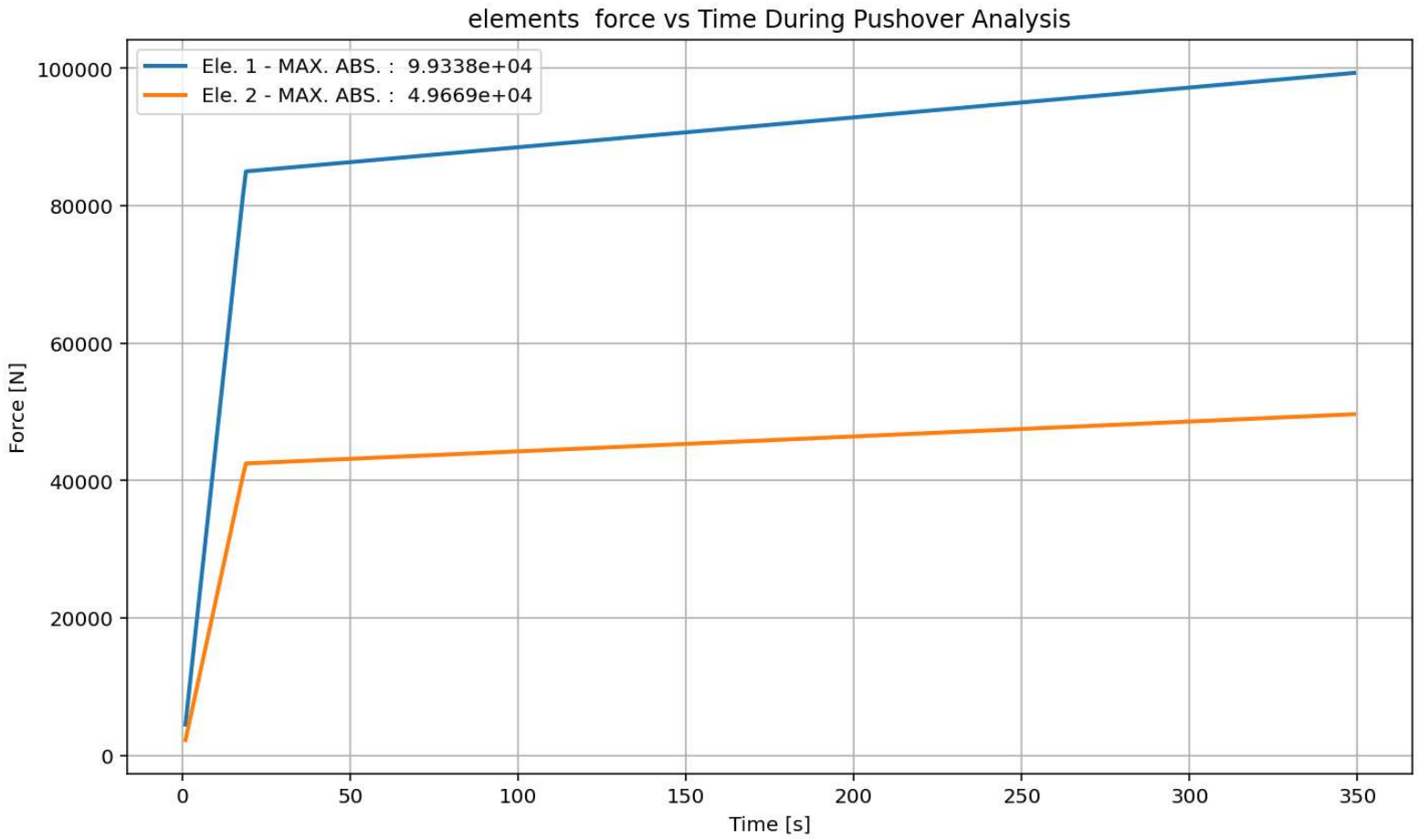
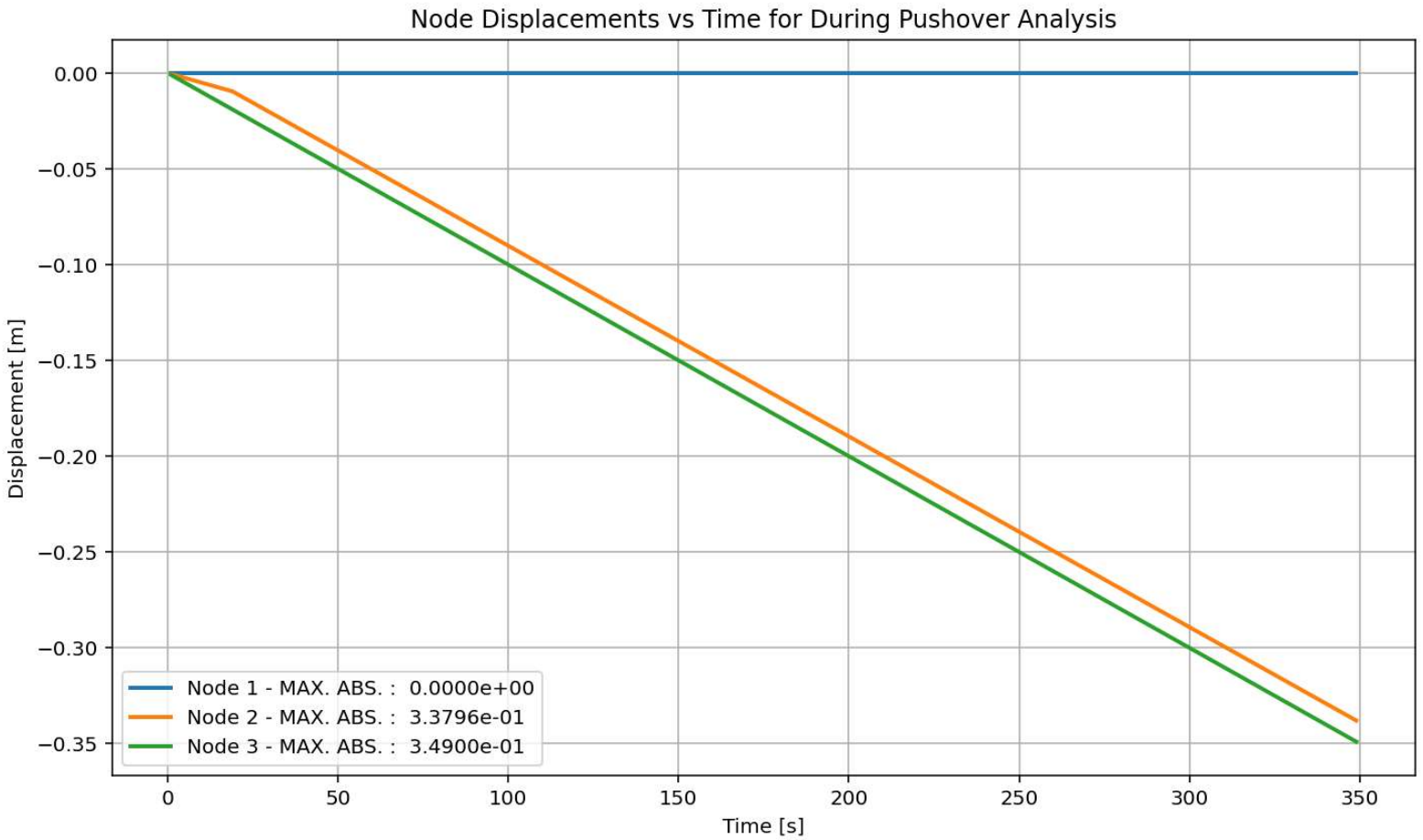
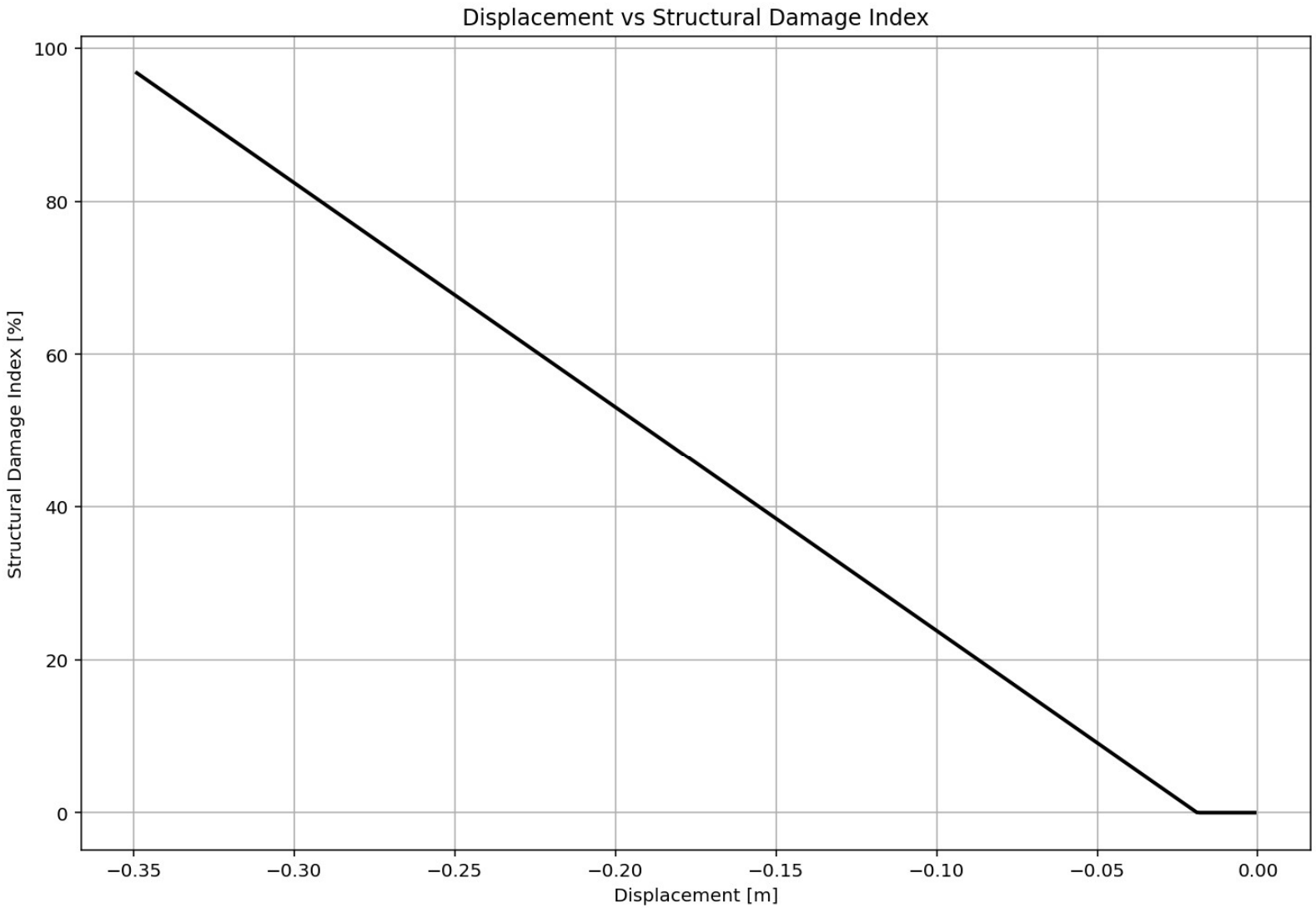

Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
100000
80000
60000
40000
20000
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]


Period of Structure During Pushover Analysis
Structural Period [s]
Displacement [m]
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.063 (s) - Max: 0.063 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 2.025 (s) - Max: 2.127 (s)
0.0
0.5
1.0
1.5
2.0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00


elements force vs Time During Pushover Analysis
Ele. 1 - MAX. ABS. : 9.9338e+04
Ele. 2 - MAX. ABS. : 4.9669e+04
Force [N]
Time [s]
0
20000
40000
60000
80000
100000
0
50
100
150
200
250
300
350


Node Displacements vs Time for During Pushover Analysis
Displacement [m]
Time [s]
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0
50
100
150
200
250
300
350
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.3796e-01
Node 3 - MAX. ABS. : 3.4900e-01


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]

Spyder (Python 3.12)

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...GY_DISSIPATION_CAPACITY_INDEX\23_SOIL_STRUCTURE_MDOF_8_ANA

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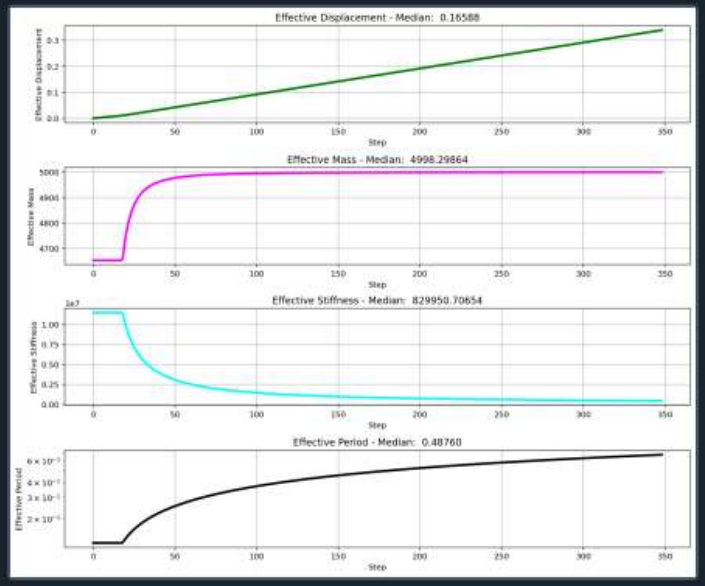
P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

```python
#%%  EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
# Change MDOF to SDOF System with Displacement Based Design Concept
# THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
This script implements a displacement-based pushover transformation,
 converting a multi-degree-of-freedom (MDOF) system into an equivalent
 single-degree-of-freedom (SDOF) system for seismic assessment.
 It calculates effective modal properties—displacement, mass, and
 stiffness—by weighting element forces and nodal displacements according
 to a presumed deformed shape.
 The derived effective period provides a simplified dynamic characteristic
 for performance-based engineering. The visualizations effectively track the
 evolution of these equivalent parameters throughout the nonlinear analysis steps.
"""
#-------------------------------------------------------------------------
displacement_X_1 = np.array(list(node_displacements.values())[1])   # DOF 02
displacement_X_2 = np.array(list(node_displacements.values())[2])   # DOF 03


ele_force_01 = np.array(list(ele_force.values())[0])   # ELEMENT 01
ele_force_02 = np.array(list(ele_force.values())[1])   # ELEMENT 02

STIFF_X_1 = np.abs(ele_force_01 / displacement_X_1)
STIFF_X_2 = np.abs(ele_force_02 / displacement_X_2)

MX2 = (MASS[0] * np.square(displacement_X_1) +
       MASS[1] * np.square(displacement_X_2))

MX = (MASS[0] * np.array(displacement_X_1) +
      MASS[1] * np.array(displacement_X_2))
EFFECTIVE_DISP_X = MX2 / np.abs(MX)

EFFECTIVE_MASS_X = np.abs(MX) / EFFECTIVE_DISP_X
```



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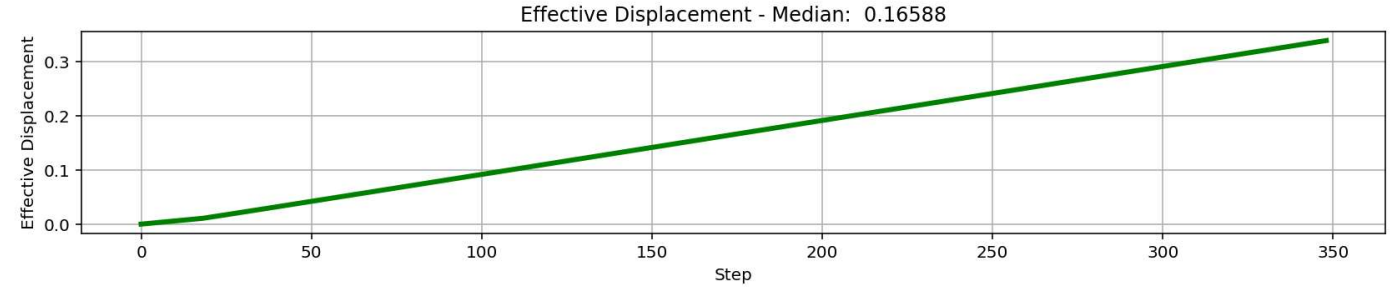
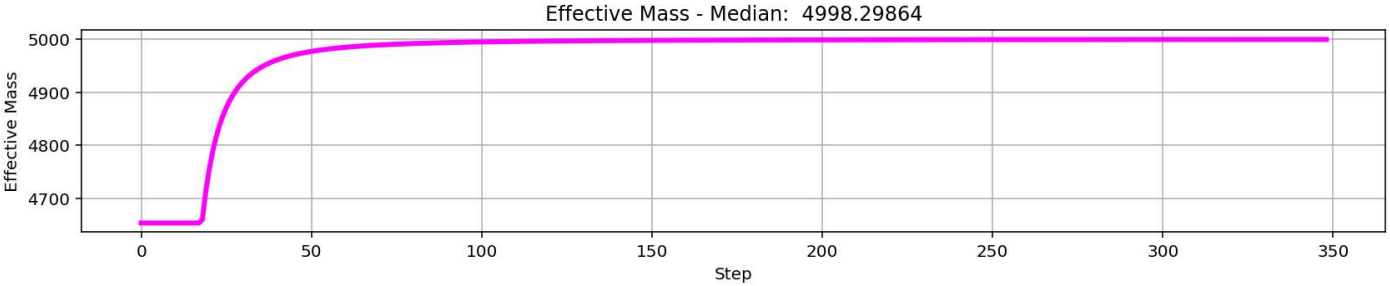
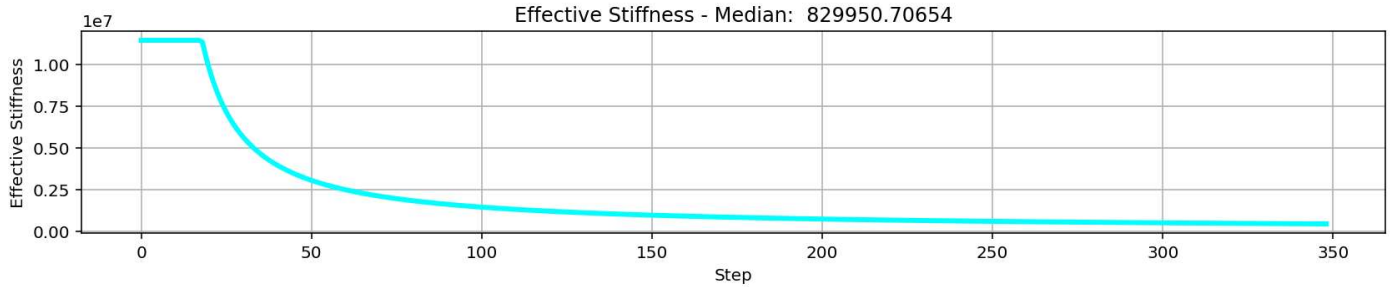


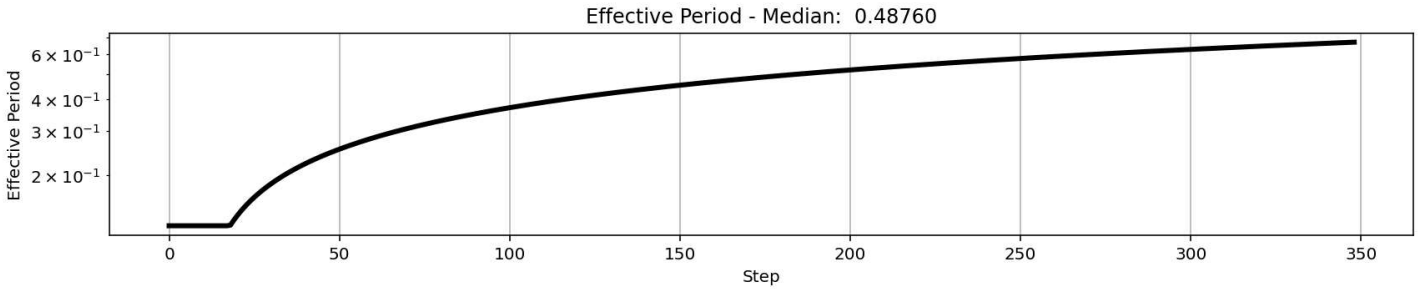
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$



$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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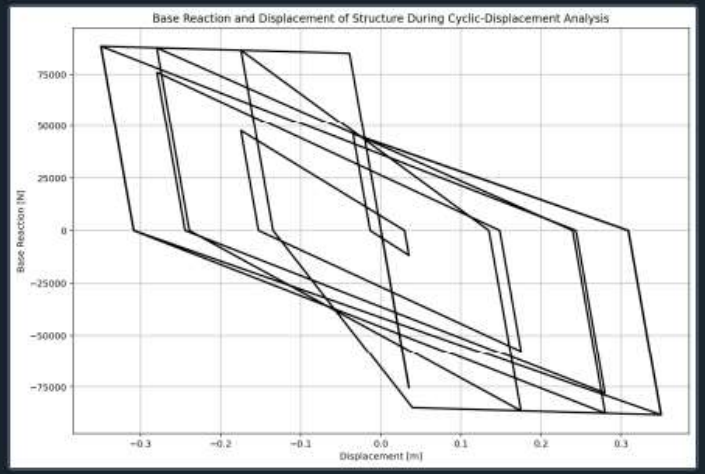
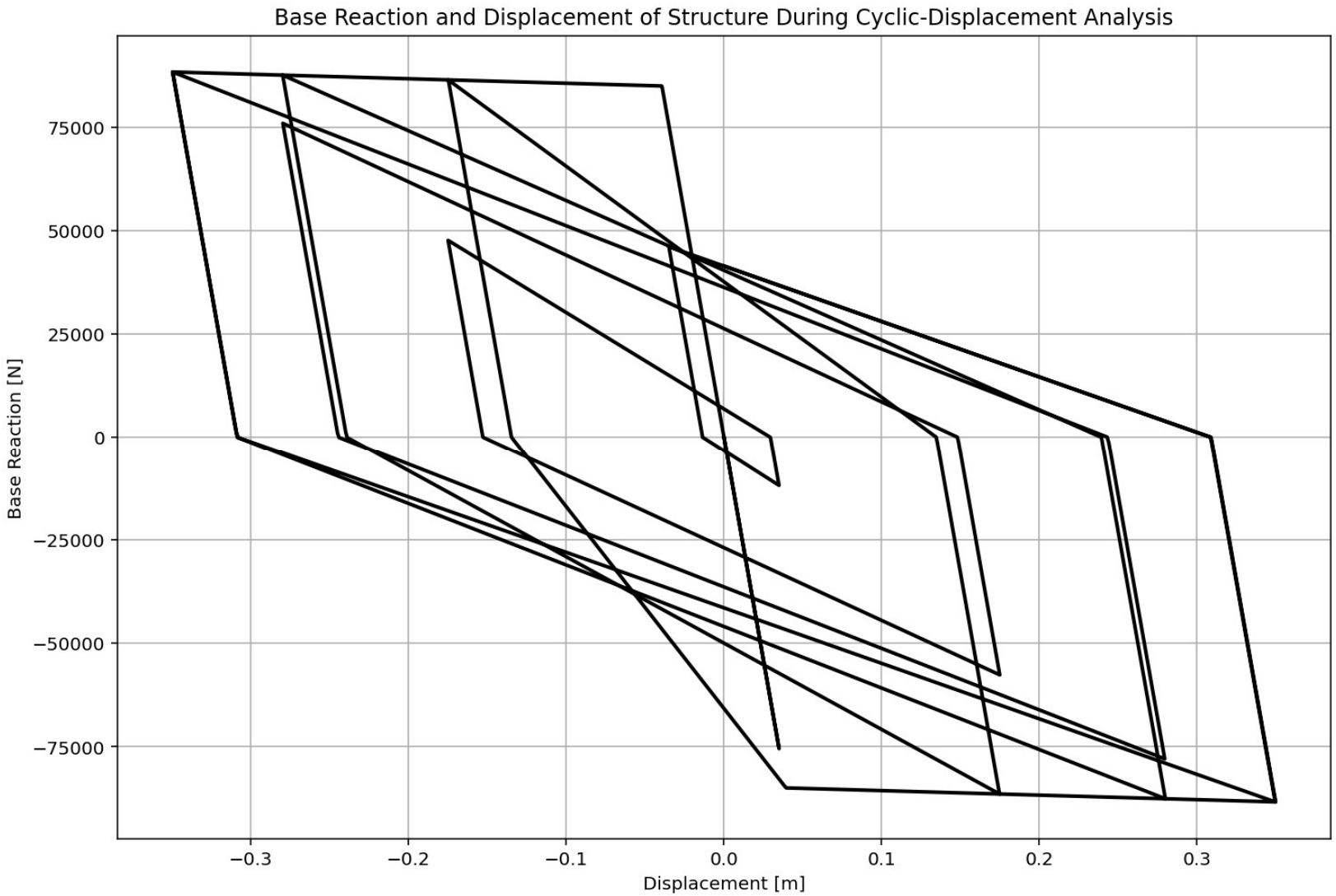
P(t)_MDOF_8_ANA.py

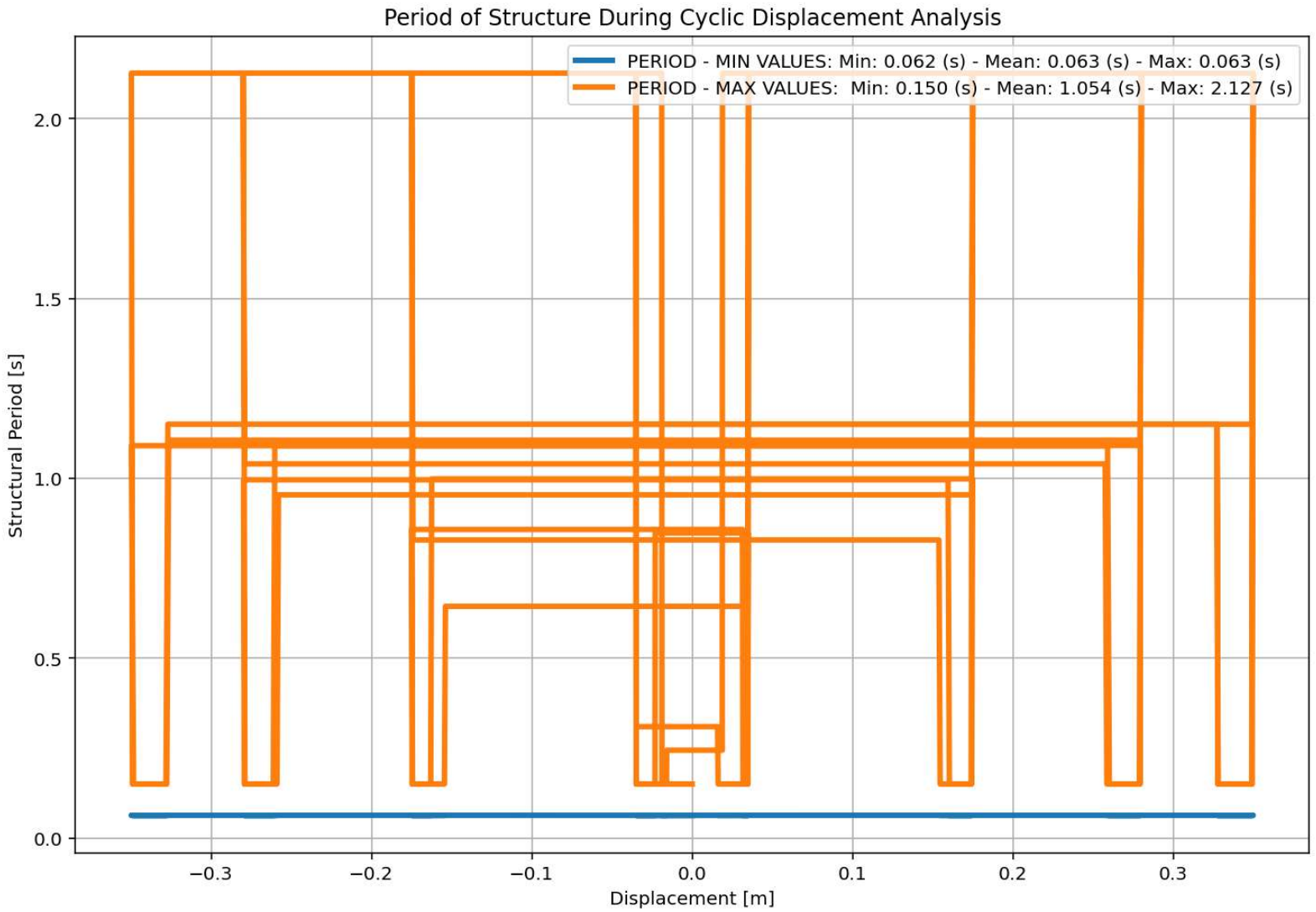
```
284     if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
285         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
286         DMAX = -0.35    # [m] Max. Displacement
287         n_points = 10000
288         # 4. CYCLIC DISPALCEMENT PROTOCOL
289         # Key strain points (same logic as your protocol)
290         key_disp = np.array([
291               0.0,
292               0.1*DMAX,   -0.1*DMAX,
293               0.5*DMAX,   -0.5*DMAX,
294               0.8*DMAX,   -0.8*DMAX,
295               DMAX,       -DMAX,
296               0.1*DMAX,     -0.1*DMAX,
297               0.5*DMAX,     -0.5*DMAX,
298               0.8*DMAX,     -0.8*DMAX,
299               DMAX,       -DMAX,
300               0.0
301         ])
302
303         # Generate 1000-point displacement protocol
304         disp_protocol = np.interp(
305             np.linspace(0, len(key_disp) - 1, n_points),
306             np.arange(len(key_disp)),
307             key_disp
308         )
309
310         ops.analysis('Static')
311         STEP = 0.0
312         for target_disp in disp_protocol:
313             current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPLIED IN MIDDLE N
314             dU = target_disp - current_disp
315             ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
316             OK = ops.analyze(1)
317             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
```

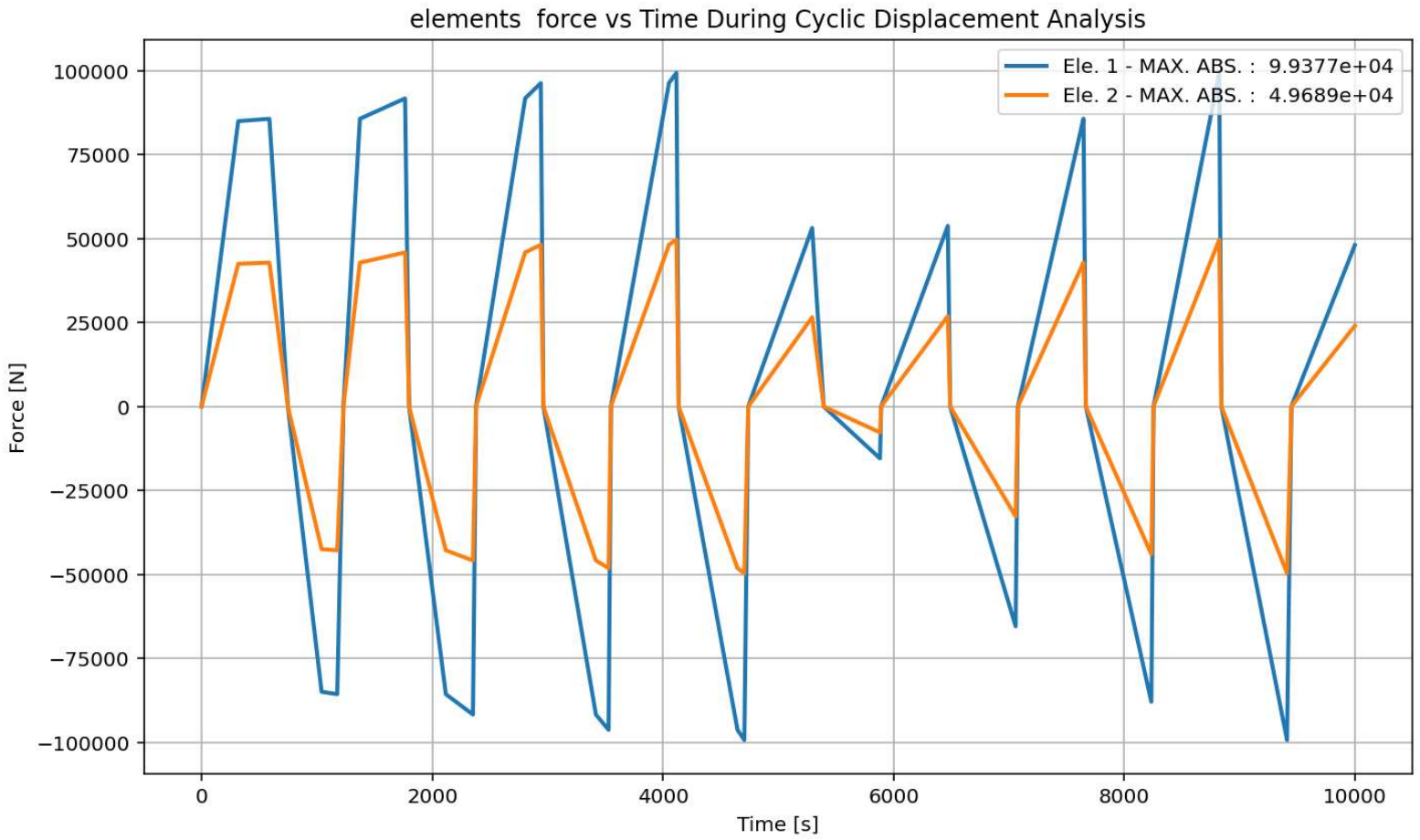


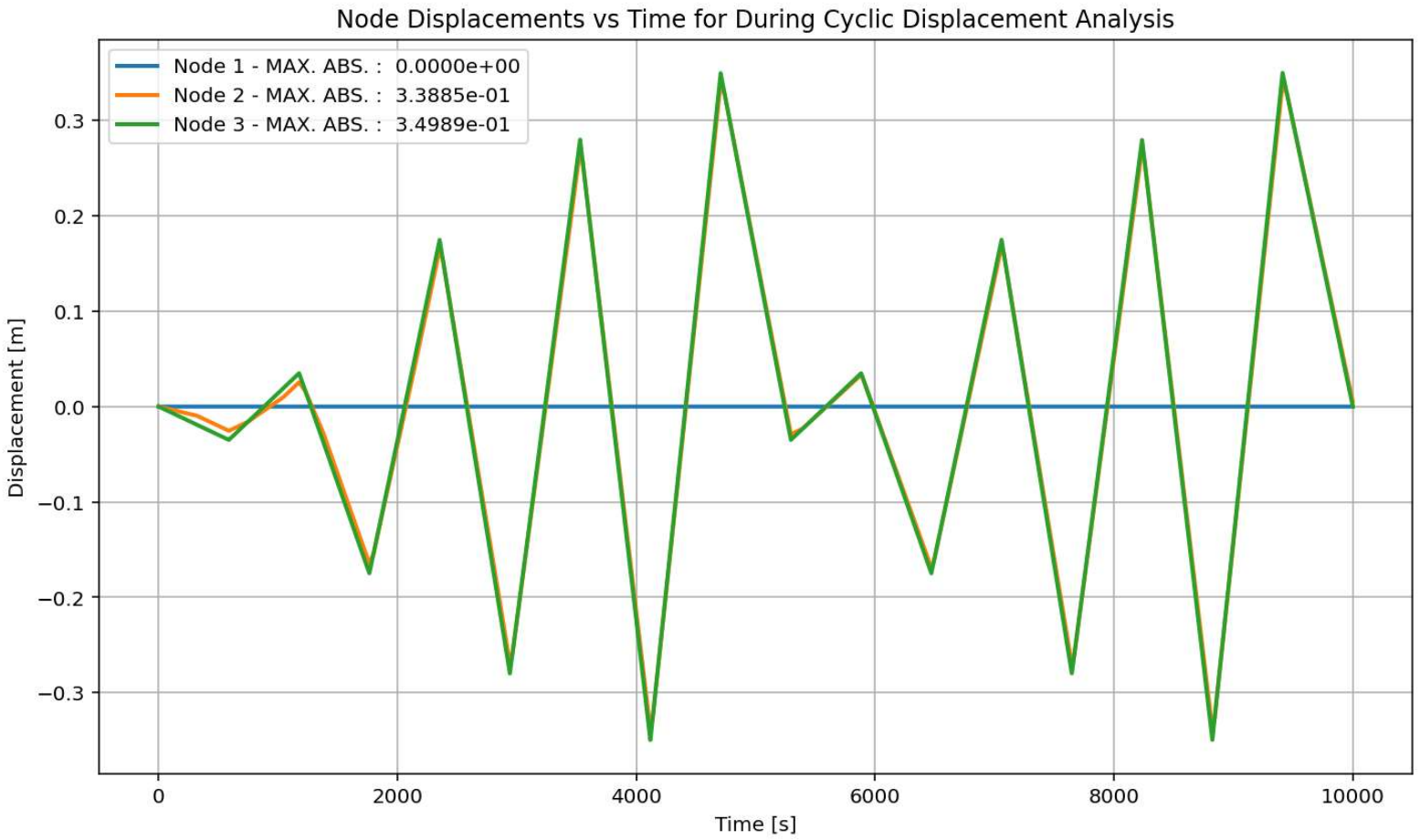
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

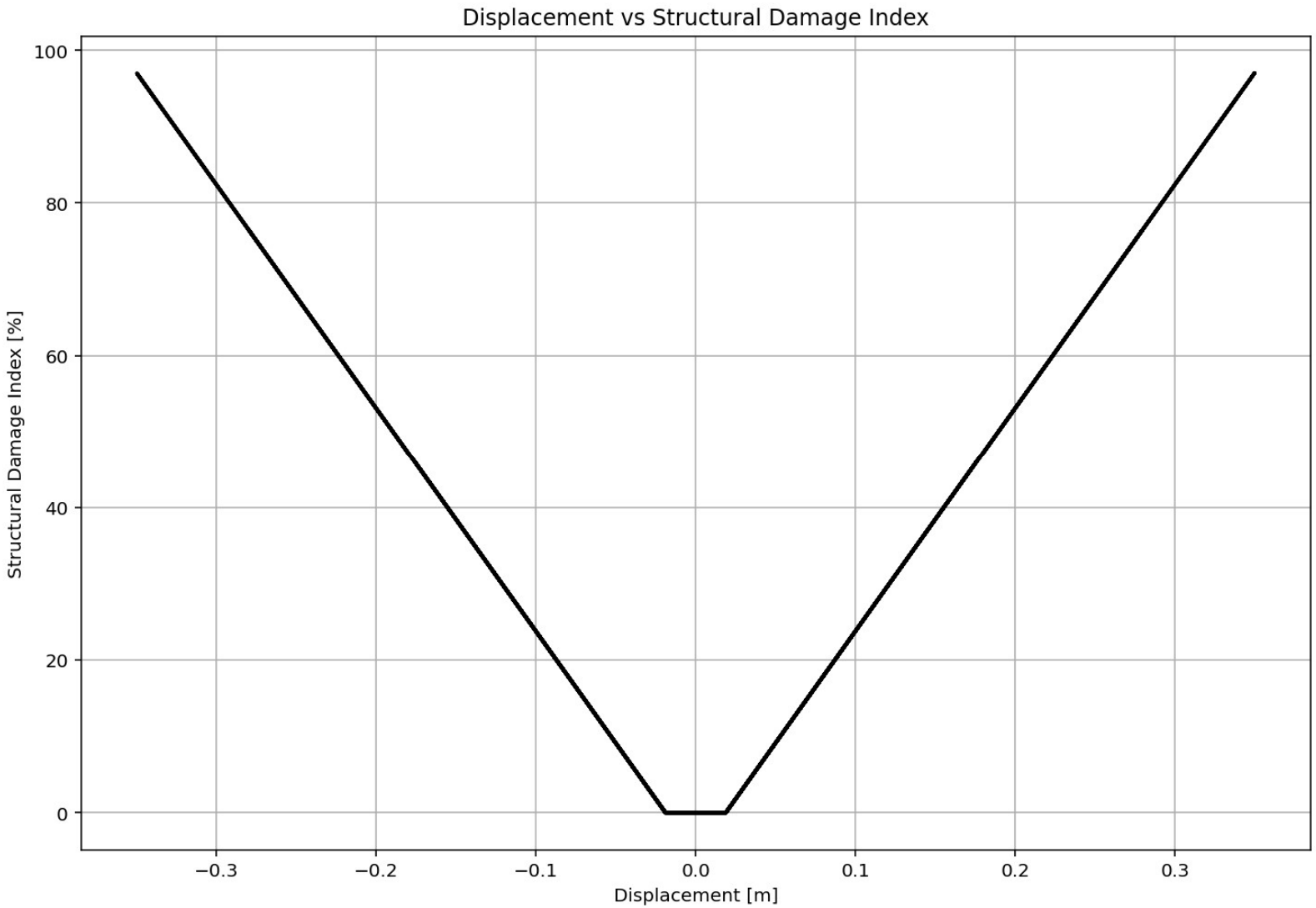
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1307, Col 23 UTF-8 CRLF RW Mem 45%

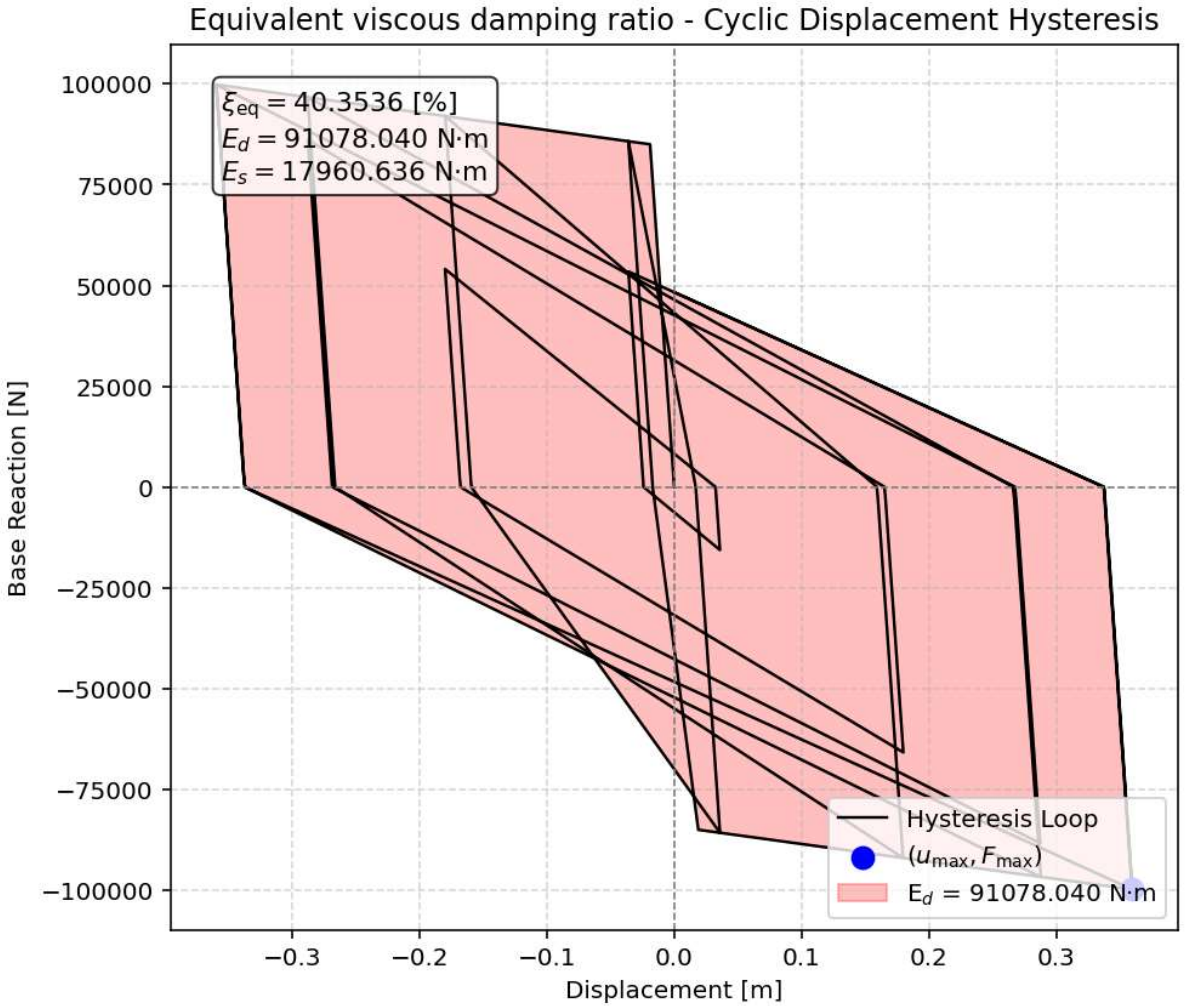

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
75000
50000
25000
0
−25000
−50000
−75000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.063 (s) - Max: 0.063 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 1.054 (s) - Max: 2.127 (s)
Structural Period [s]
2.0
1.5
1.0
0.5
0.0
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


elements force vs Time During Cyclic Displacement Analysis
Ele. 1 - MAX. ABS. : 9.9377e+04
Ele. 2 - MAX. ABS. : 4.9689e+04
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0
2000
4000
6000
8000
10000
Time [s]


Node Displacements vs Time for During Cyclic Displacement Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.3885e-01
Node 3 - MAX. ABS. : 3.4989e-01
Displacement [m]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
2000
4000
6000
8000
10000
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 40.3536$ [%]
$E_d = 91078.040$ N·m
$E_s = 17960.636$ N·m
Base Reaction [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 91078.040 N·m

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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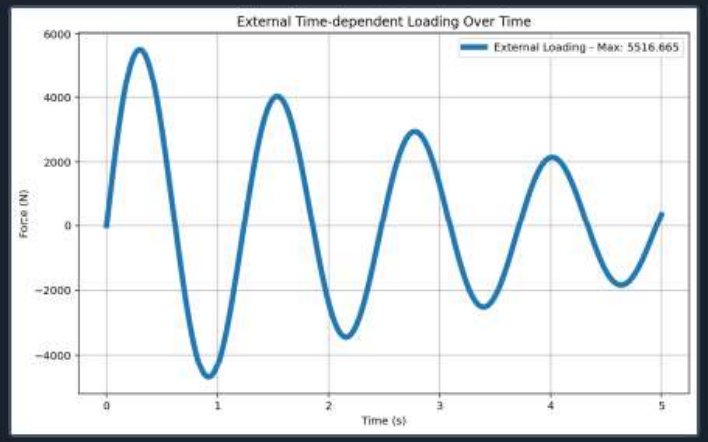
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...N_CAPACITY_INDEX\22_MDOF_8_ANA\P(t)_MDOF_8_ANA.py

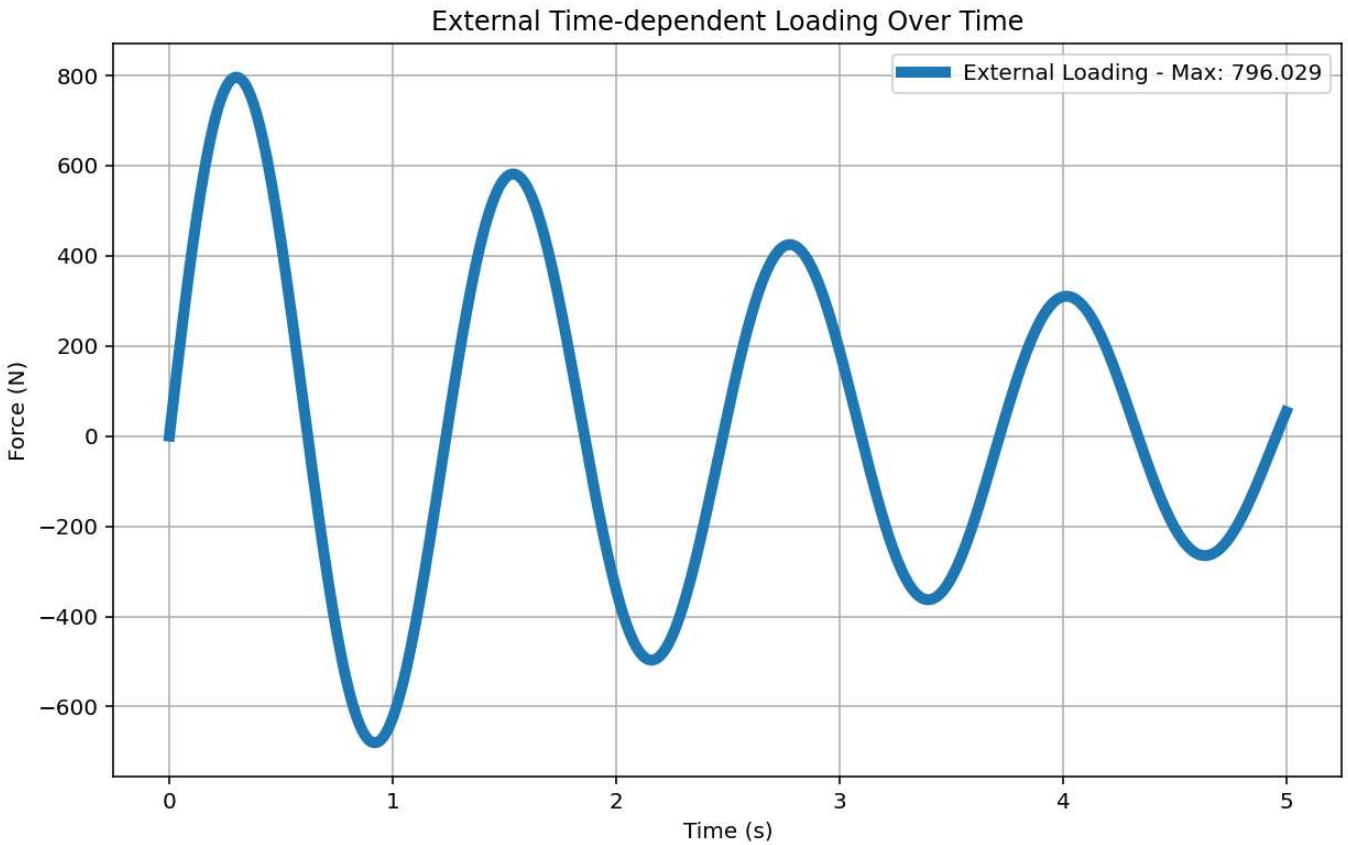
P(t)_MDOF_8_ANA.py

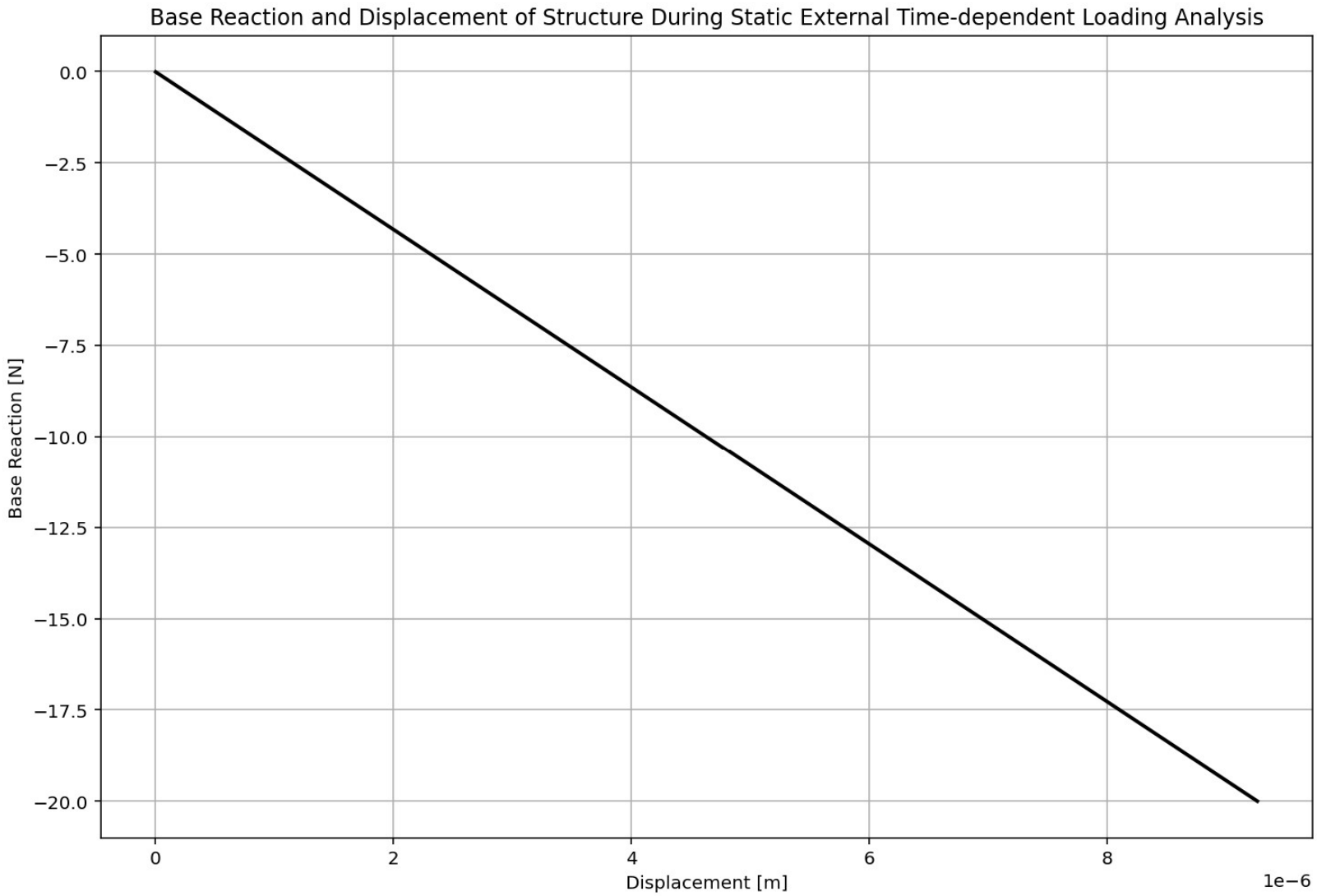
```
349     if ANAL_TYPE == 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING': # STATIC TIME-HISTORY ANALYS
350         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
351         # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
352         duration = 20.0             # [s] Analysis duration
353         dt = 0.01                   # [s] Time step
354         DT = dt                     # [s] Time step
355         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
356         force_amplitude = 5960.0    # [N] Amplitude Force
357         omega_DT = 5.0715           # [rad/s] Natural angular frequency
358         time_steps = int(duration/dt)
359
360         # Check function
361         def CHECK_FUN(DT_time ,duration):
362             if DT_time > duration:
363                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n
364                 exit()
365             return -1
366
367         CHECK_FUN(DT_time ,duration)
368         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
369             import numpy as np
370             import matplotlib.pyplot as plt
371             # External Load Durations
372             num_steps = int(DT_time / DT)
373             load_time = np.linspace(0, DT_time, num_steps)
374             target_frequency = 1.0 * omega_DT  # Target excitation frequency
375             DT_load = force_amplitude * np.sin(target_frequency * load_time)
376             # Plot External Time-dependent Loading
377             plt.figure(figsize=(10, 6))
378             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):
379             plt.title('External Time-dependent Loading Over Time')
380             plt.xlabel('Time (s)')
381             plt.ylabel('Force (N)')
382             plt.grid(True)
```

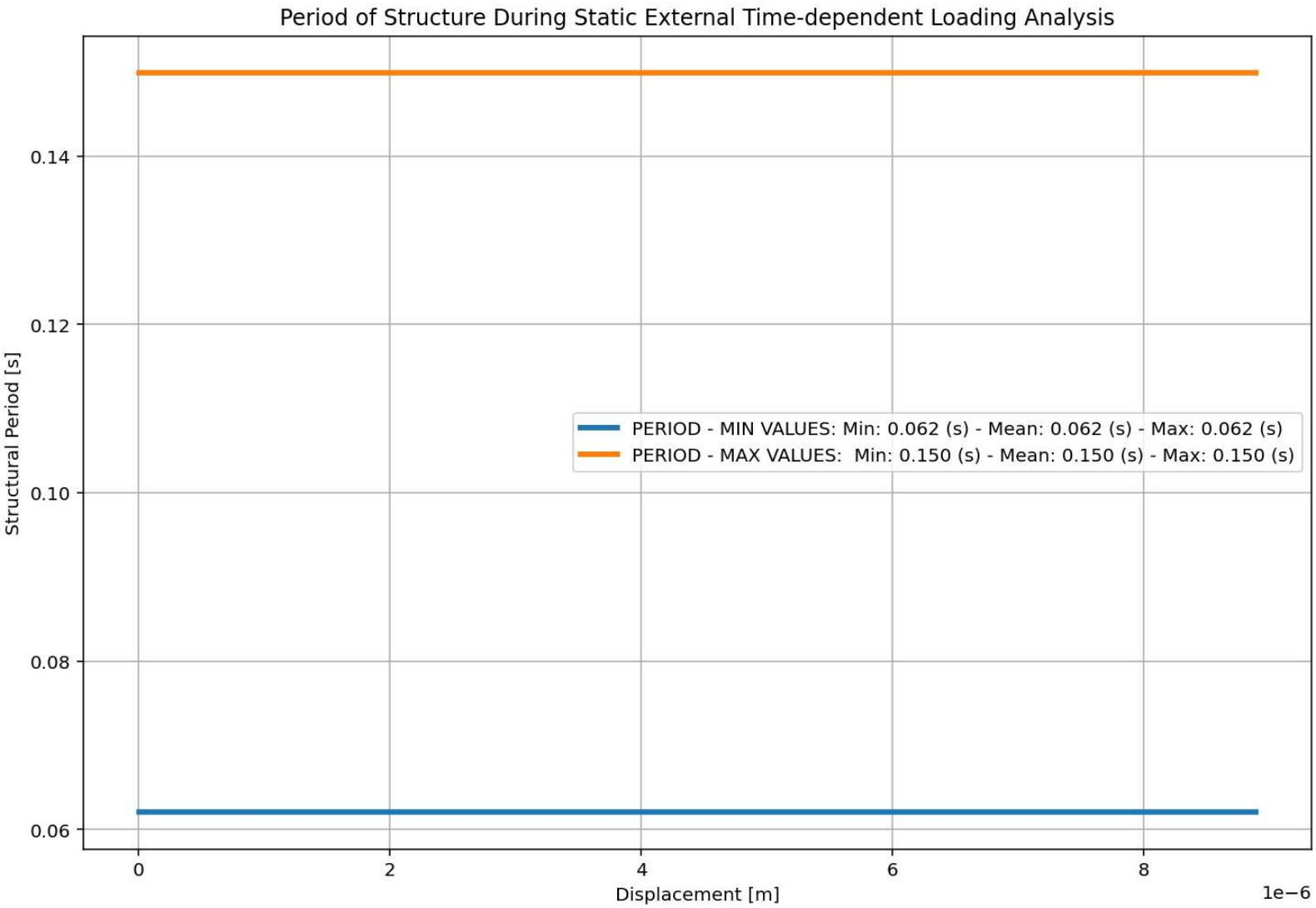


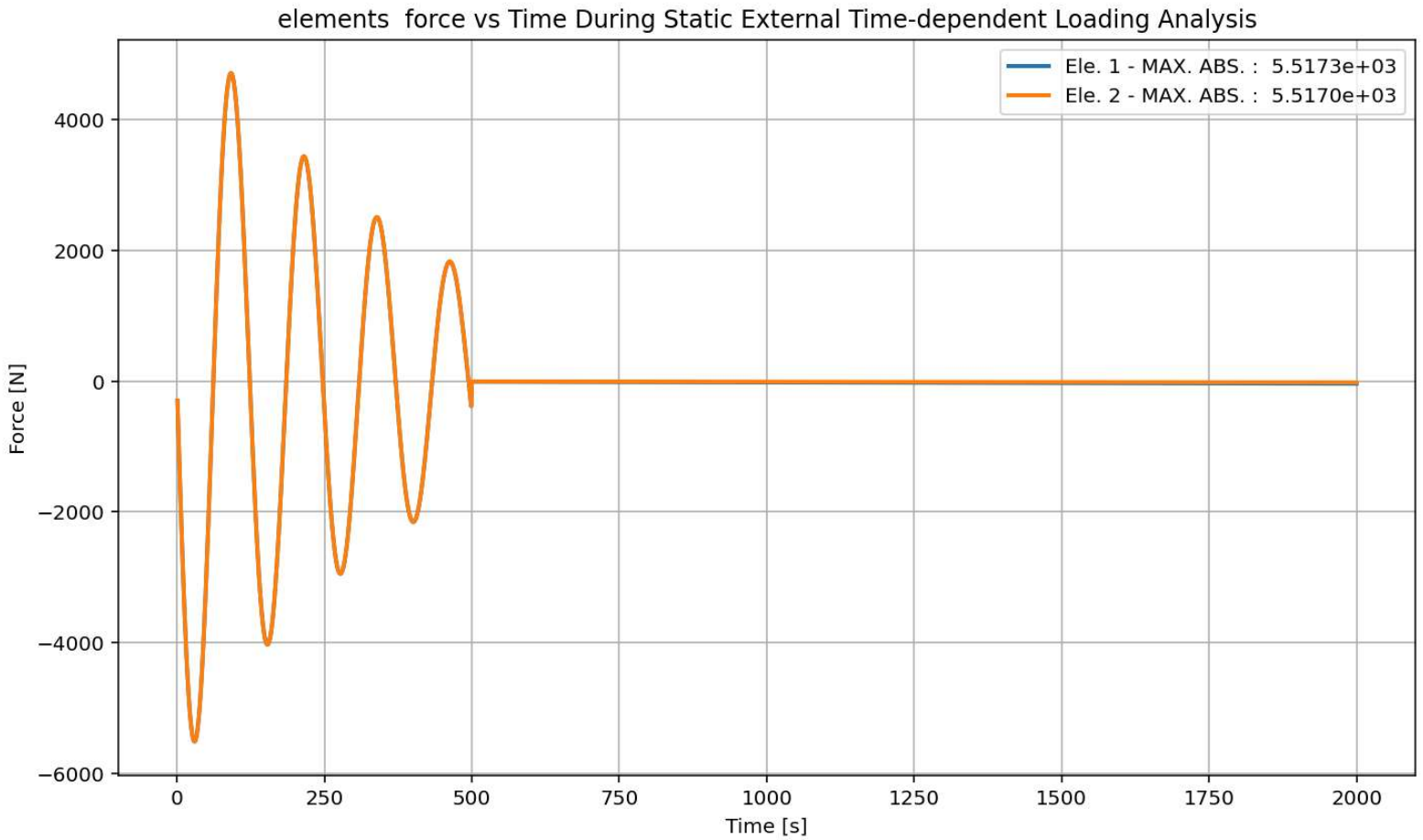
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

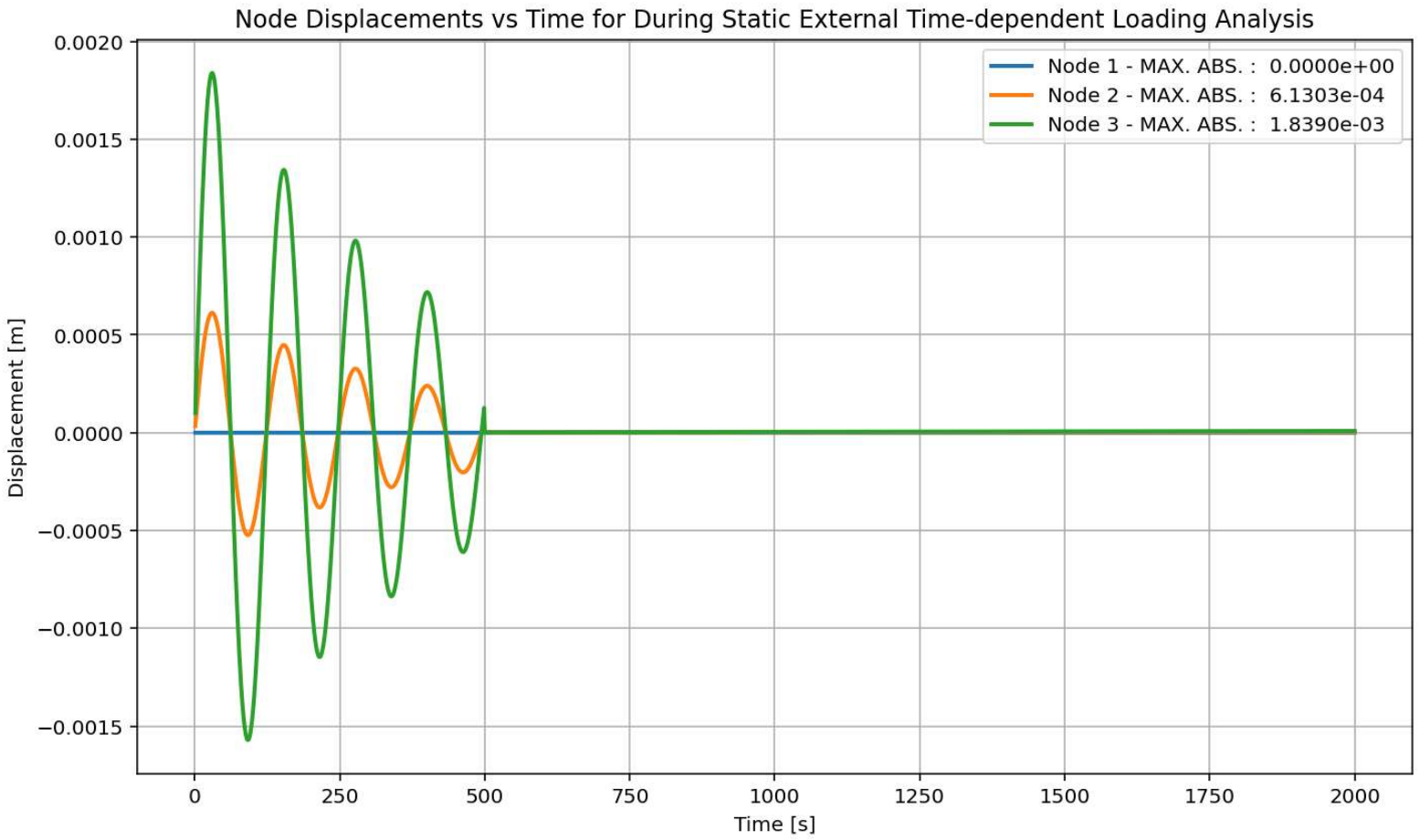
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1307, Col 23 UTF-8 CRLF RW Mem 45%

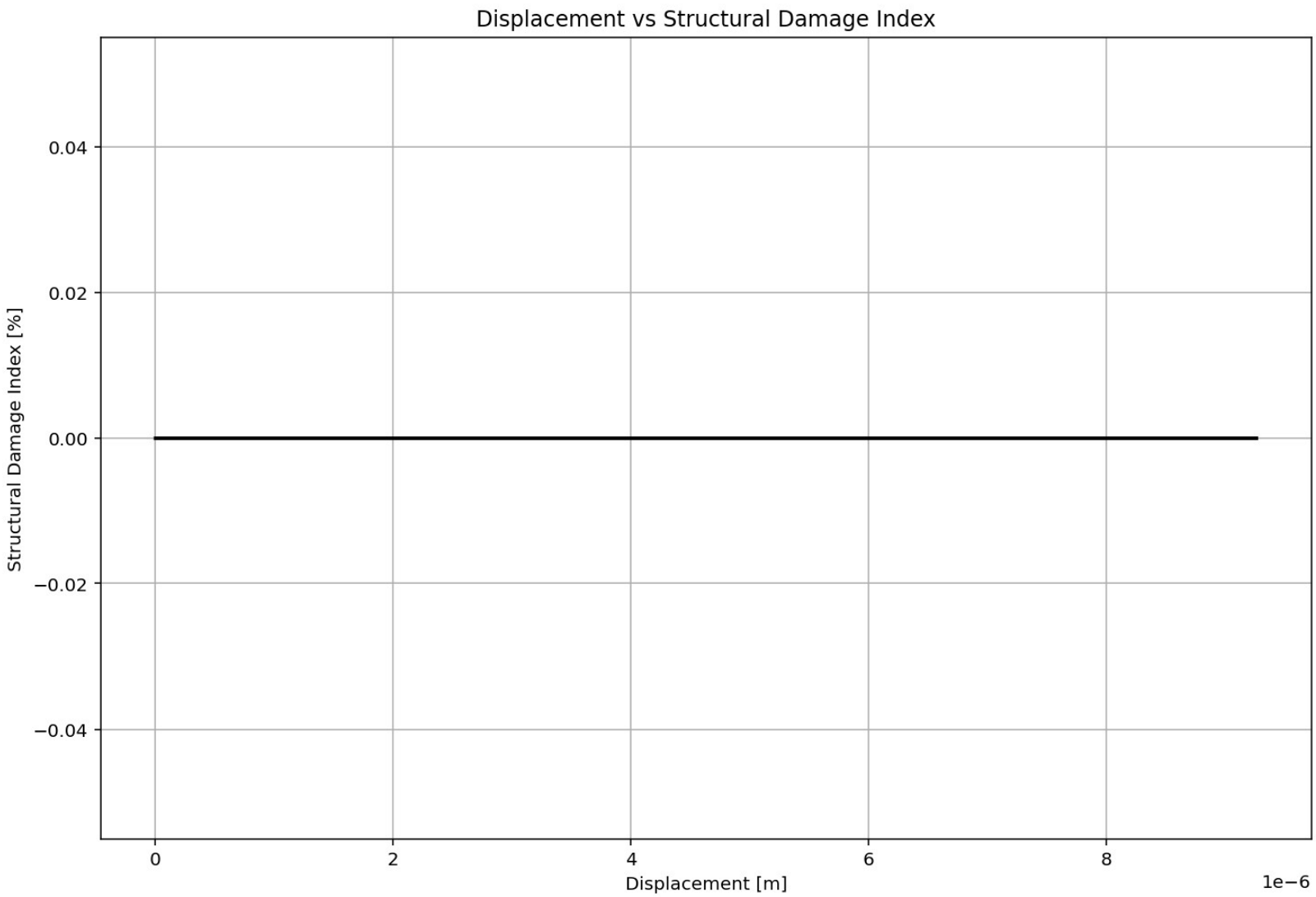

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
0
2
4
6
8
Displacement [m]
1e−6


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.14
0.12
0.10
0.08
0.06
0
2
4
6
8
Displacement [m]
1e−6
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.062 (s) - Max: 0.062 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 0.150 (s) - Max: 0.150 (s)


elements force vs Time During Static External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 5.5173e+03
Ele. 2 - MAX. ABS. : 5.5170e+03
Force [N]
4000
2000
0
−2000
−4000
−6000
0
250
500
750
1000
1250
1500
1750
2000
Time [s]


Node Displacements vs Time for During Static External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.1303e-04
Node 3 - MAX. ABS. : 1.8390e-03
Displacement [m]
Time [s]
0.0020
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
0
250
500
750
1000
1250
1500
1750
2000


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
0
2
4
6
8
Displacement [m]
1e−6

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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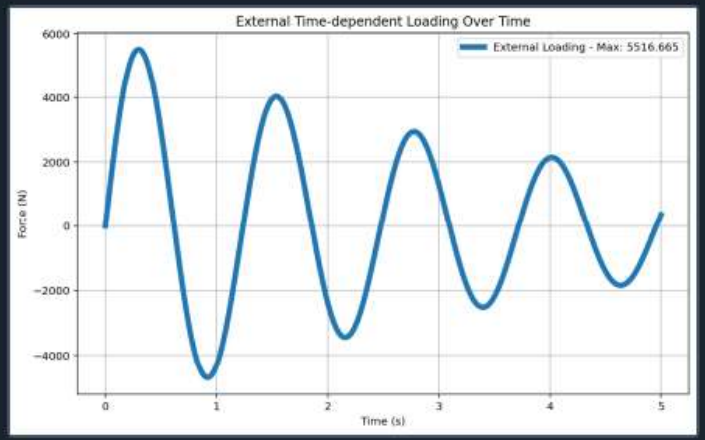
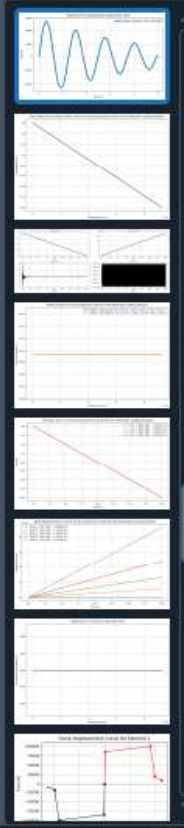
...GE_INDEX_&_ENERGY_DISSIPATION_CAPACITY_INDEX\22_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...N_CAPACITY_INDEX\22_MDOF_8_ANA\P(t)_MDOF_8_ANA.py

P(t)_MDOF_8_ANA.py

```
462     if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TIME-HISTORY ANAL
463         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
464         duration = 20.0             # [s] Analysis duration
465         dt = 0.01                   # [s] Time step
466         DT = dt                     # [s] Time step
467         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
468         force_amplitude = 5960.0    # [N] Amplitude Force
469         omega_DT = 5.0715           # [rad/s] Natural angular frequency
470         DR = 0.03                   # Damping Ratio
471
472         # Check function
473         def CHECK_FUN(DT_time ,duration):
474             if DT_time > duration:
475                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n
476                 exit()
477             return -1
478
479         CHECK_FUN(DT_time ,duration)
480         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
481             import numpy as np
482             import matplotlib.pyplot as plt
483             # External Load Durations
484             num_steps = int(DT_time / DT)
485             load_time = np.linspace(0, DT_time, num_steps)
486             target_frequency = 1.0 * omega_DT  # Target excitation frequency
487             DT_load = force_amplitude * np.sin(target_frequency * load_time)
488             # Plot External Time-dependent Loading
489             plt.figure(figsize=(10, 6))
490             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):
491             plt.title('External Time-dependent Loading Over Time')
492             plt.xlabel('Time (s)')
493             plt.ylabel('Force (N)')
494             plt.grid(True)
495             plt.legend()
```

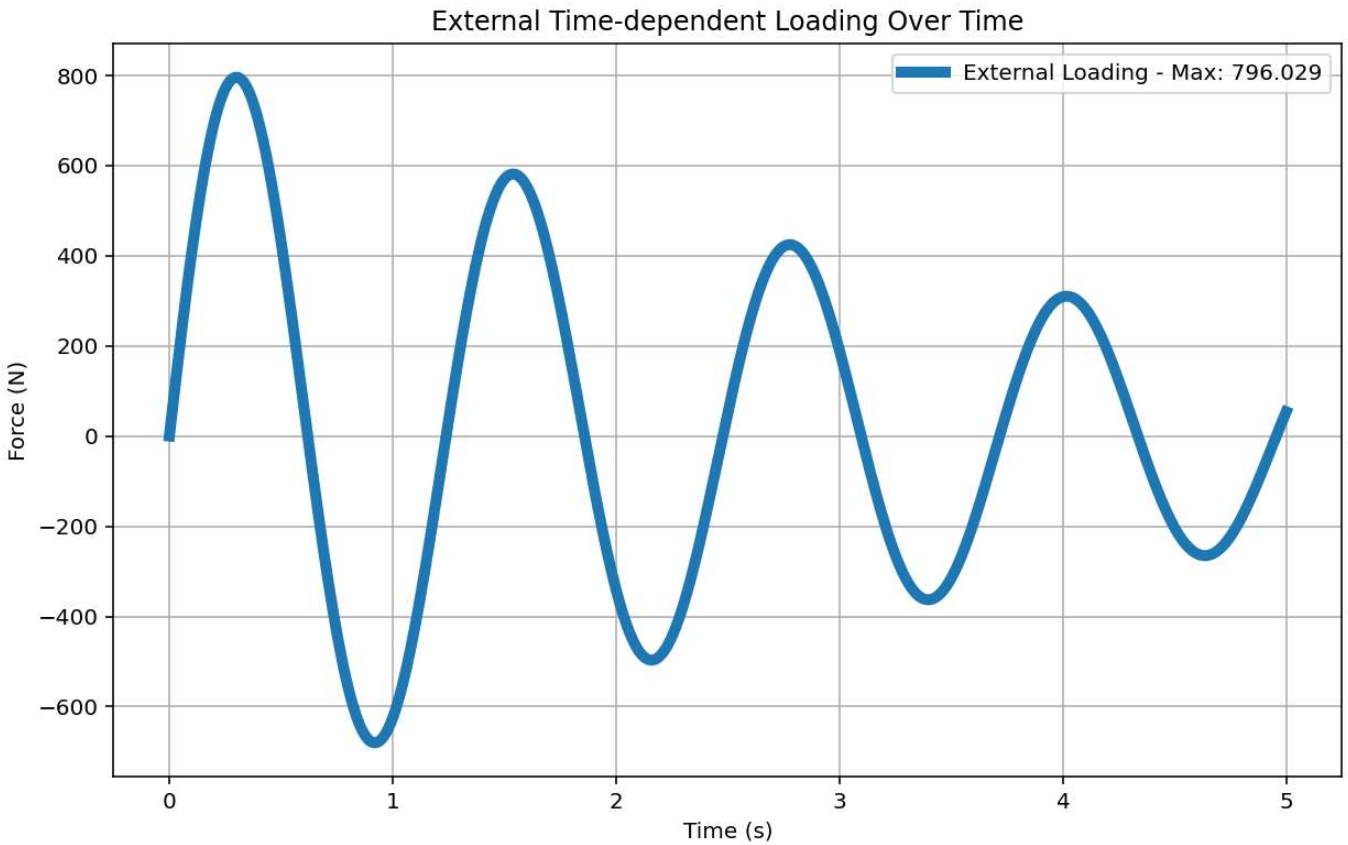


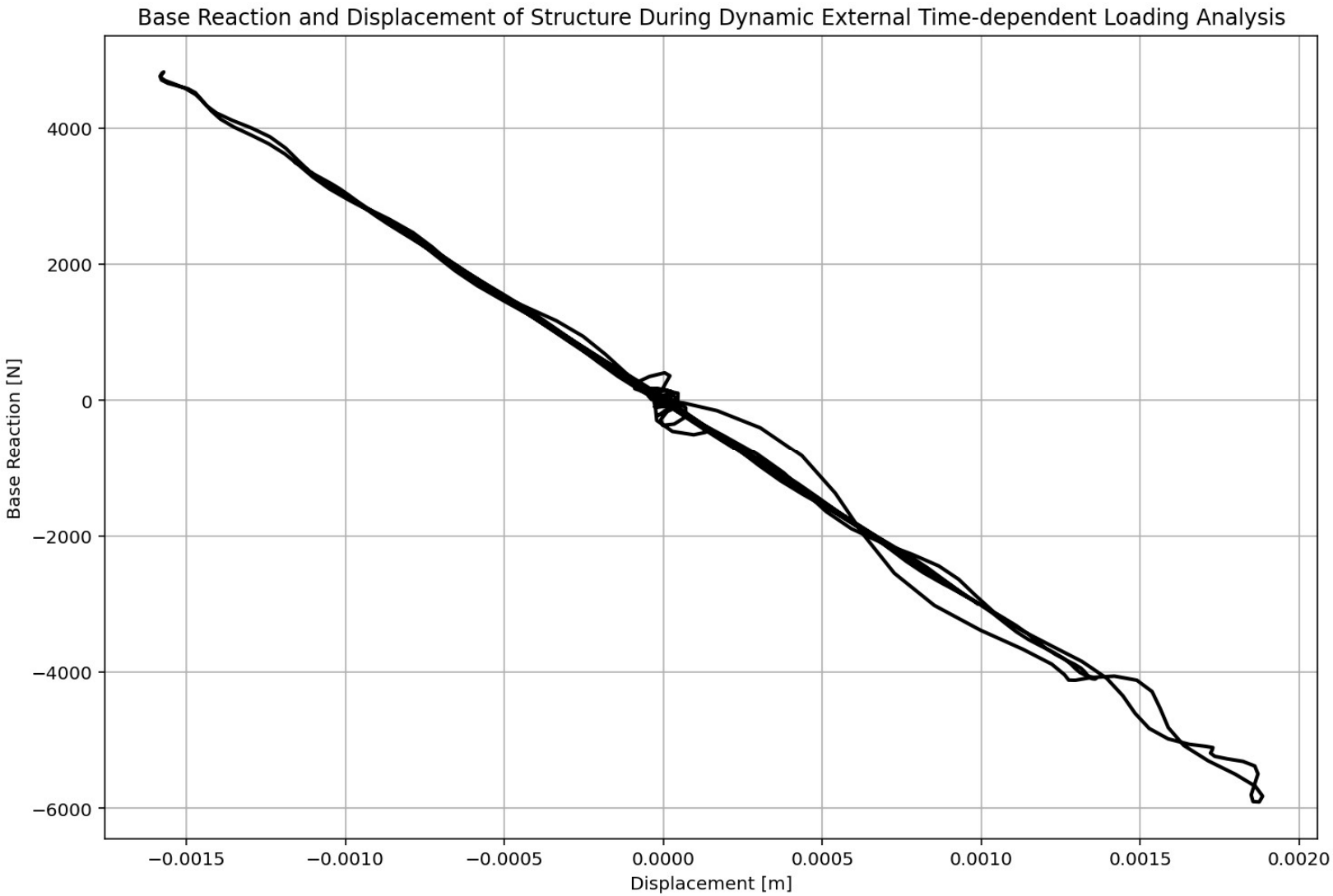


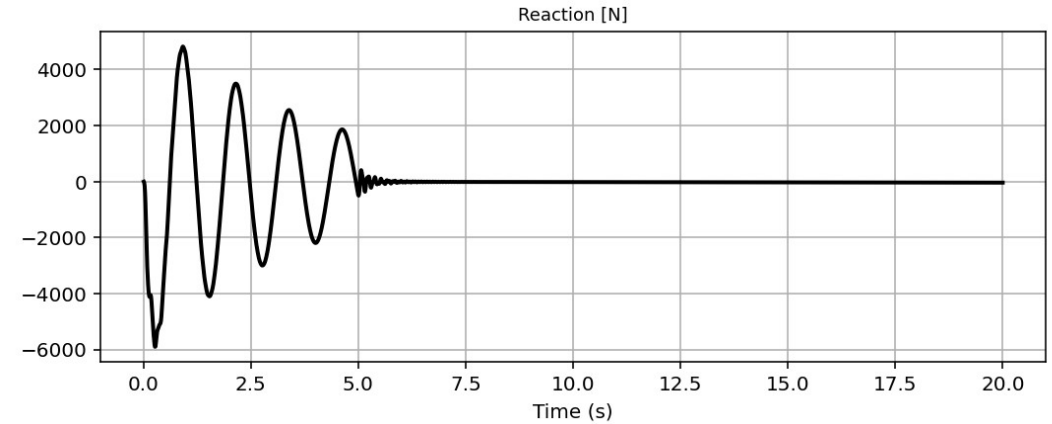
36 %

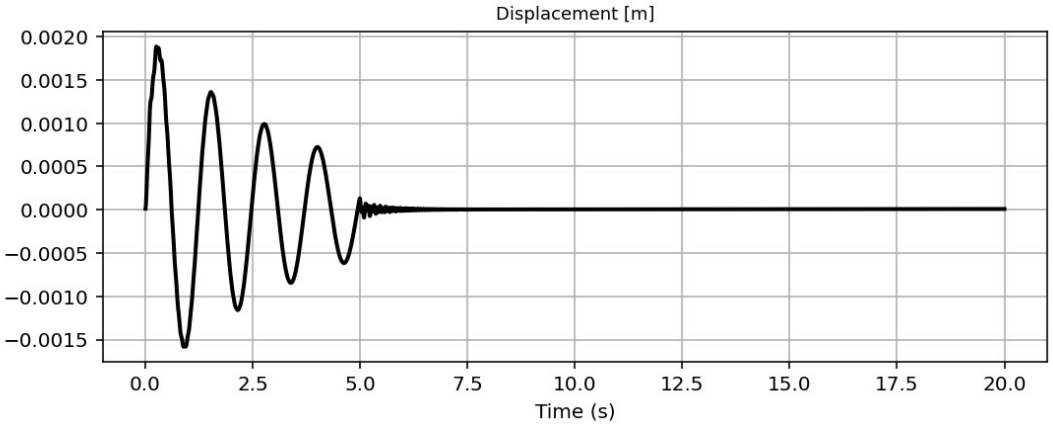
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

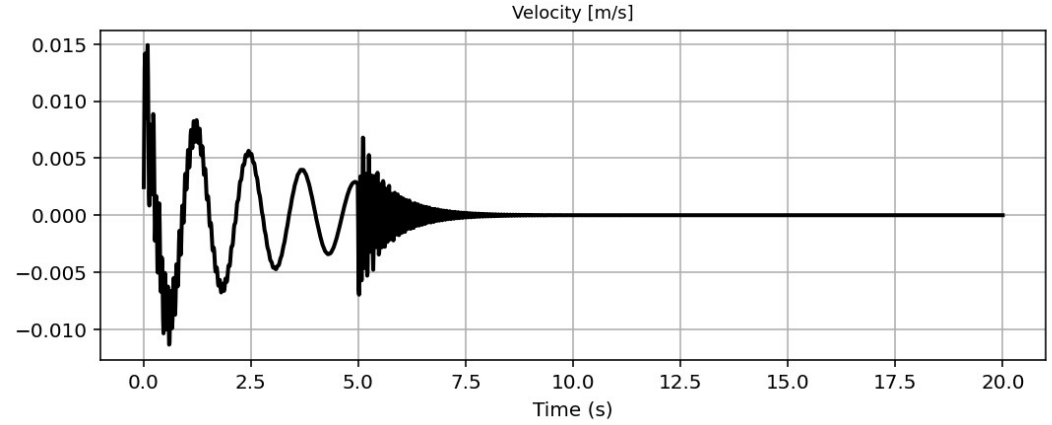
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1307, Col 23 UTF-8 CRLF RW Mem 46%

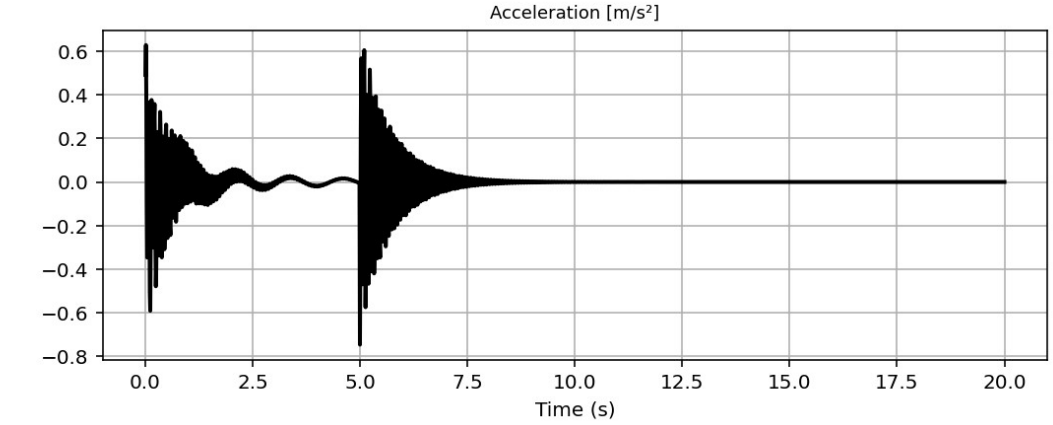

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)

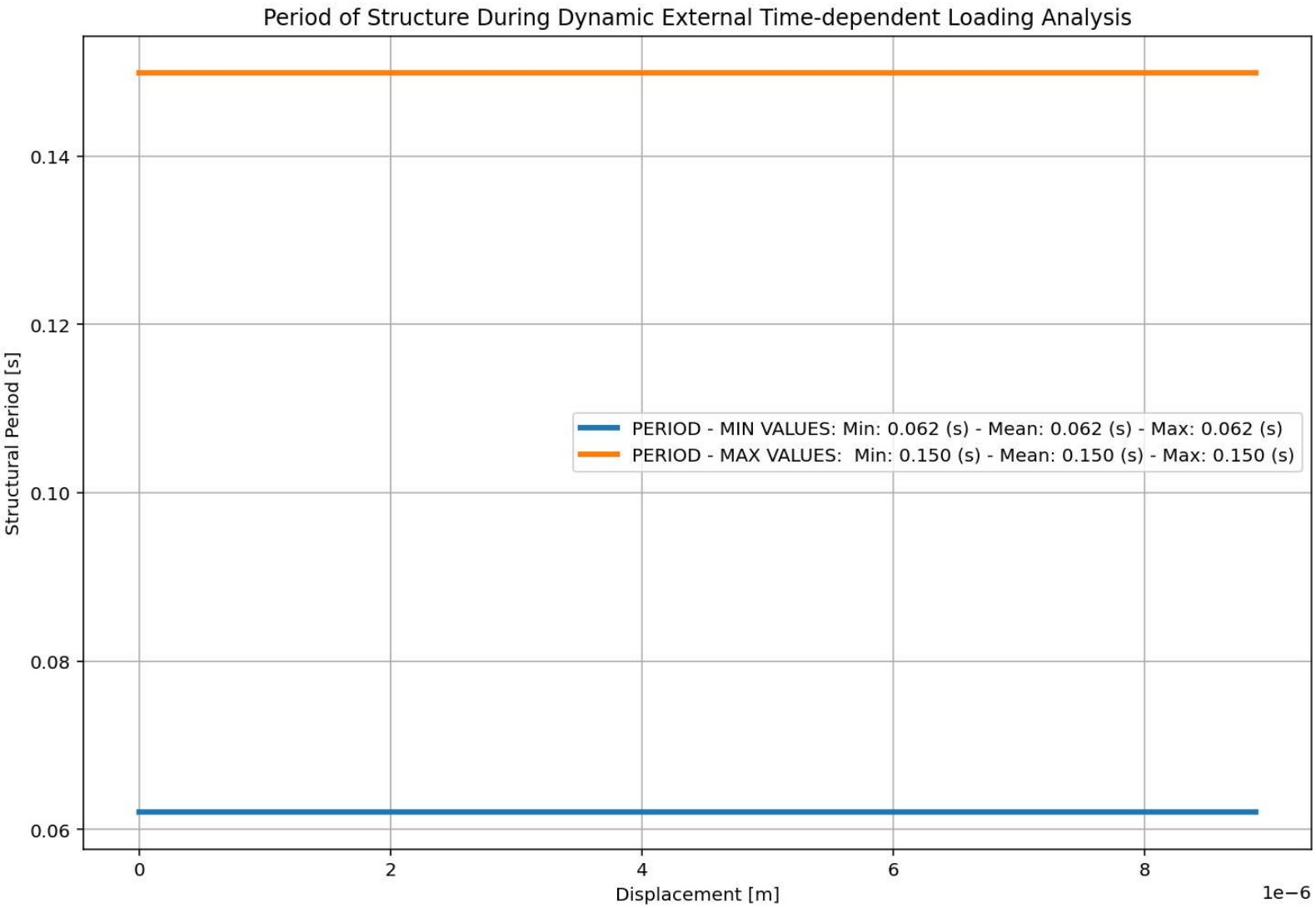

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
4000
2000
0
−2000
−4000
−6000
−0.0015
−0.0010
−0.0005
0.0000
0.0005
0.0010
0.0015
0.0020
Displacement [m]

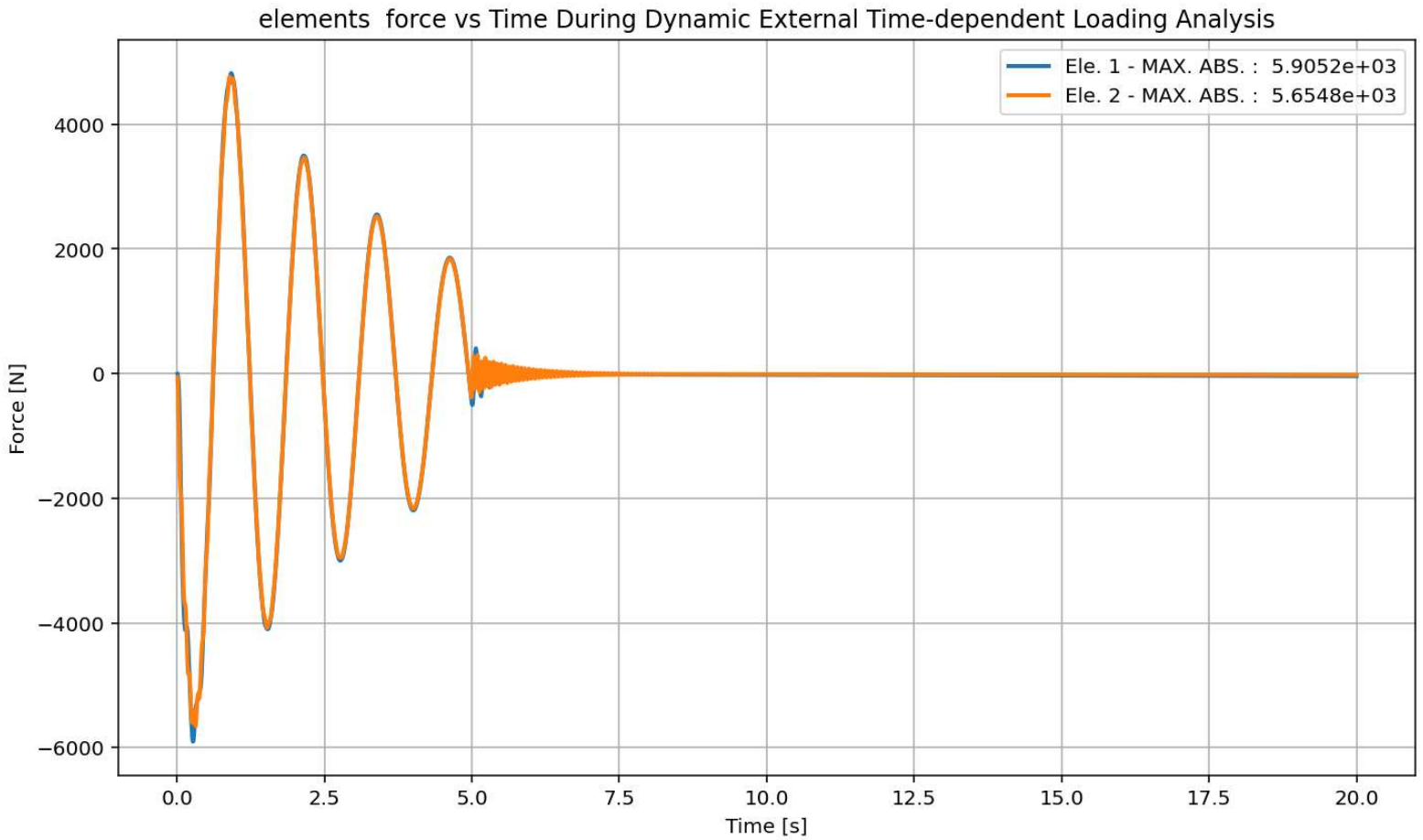

Reaction [N]
4000
2000
0
−2000
−4000
−6000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

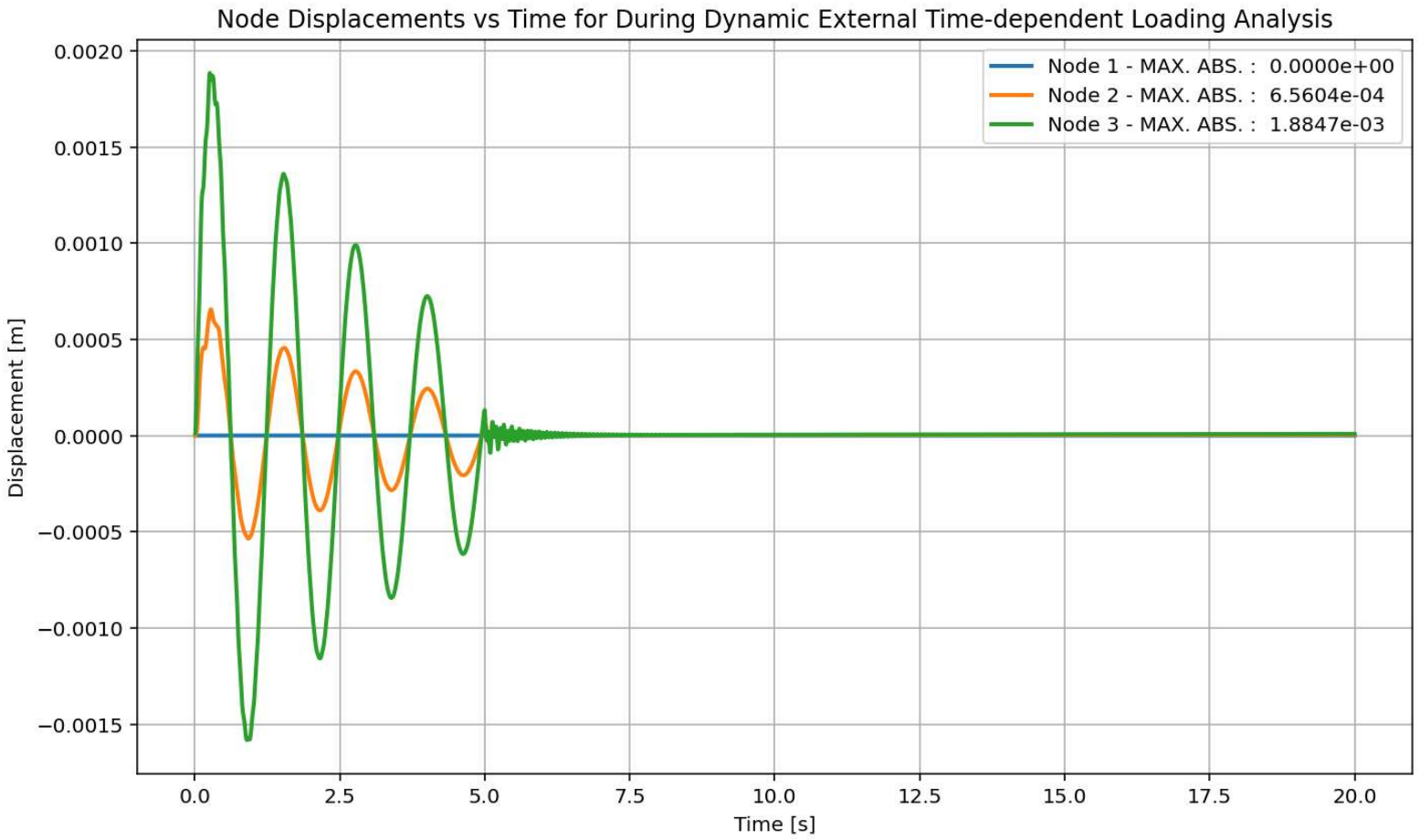

Displacement [m]
0.0020
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

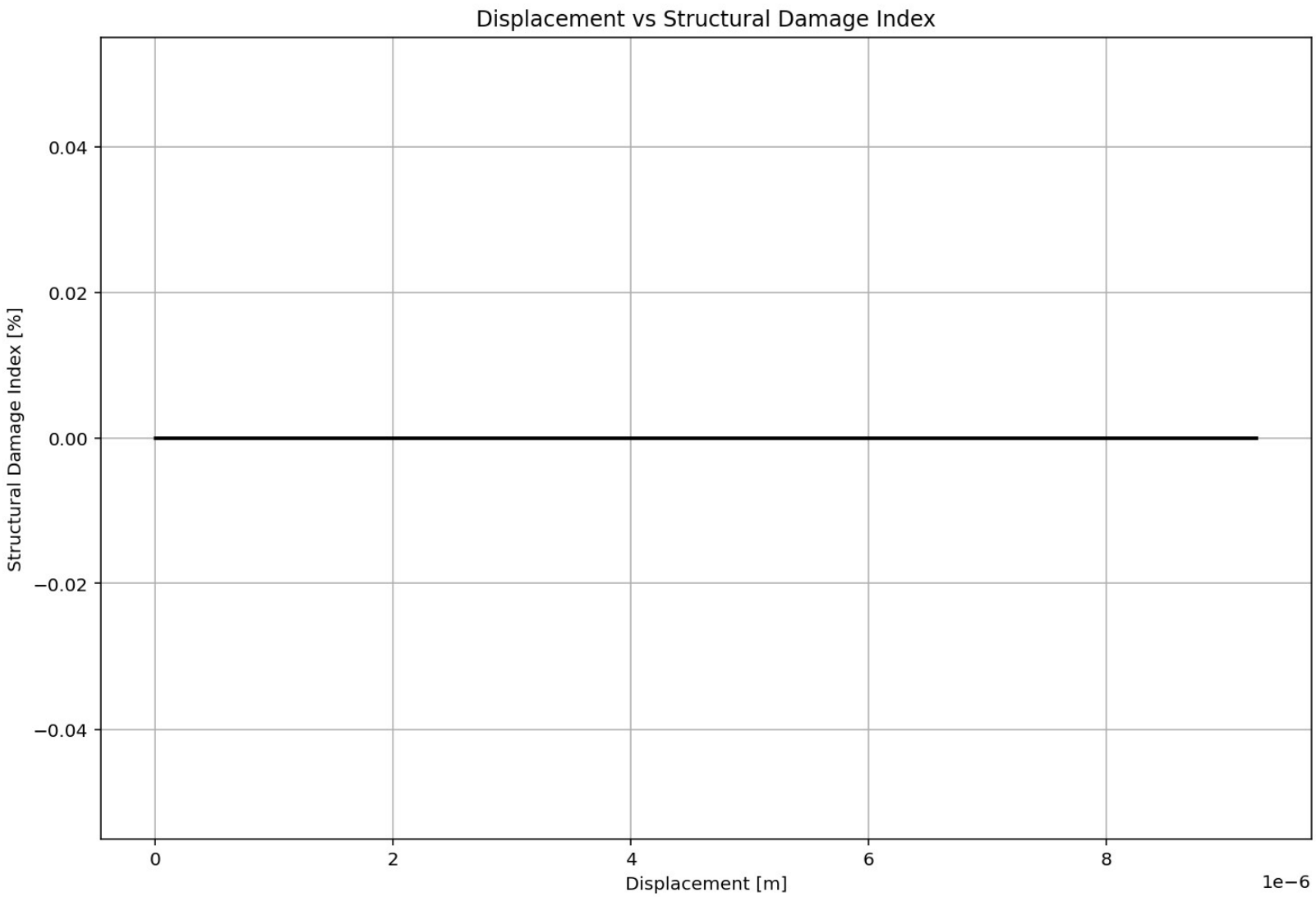

Velocity [m/s]
0.015
0.010
0.005
0.000
−0.005
−0.010
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Acceleration [m/s²]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
0.14
0.12
0.10
0.08
0.06
0
2
4
6
8
Displacement [m]
1e−6
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.062 (s) - Max: 0.062 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 0.150 (s) - Max: 0.150 (s)


elements force vs Time During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 5.9052e+03
Ele. 2 - MAX. ABS. : 5.6548e+03
Force [N]
4000
2000
0
−2000
−4000
−6000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.5604e-04
Node 3 - MAX. ABS. : 1.8847e-03
Displacement [m]
0.0020
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
0
2
4
6
8
Displacement [m]
1e−6

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

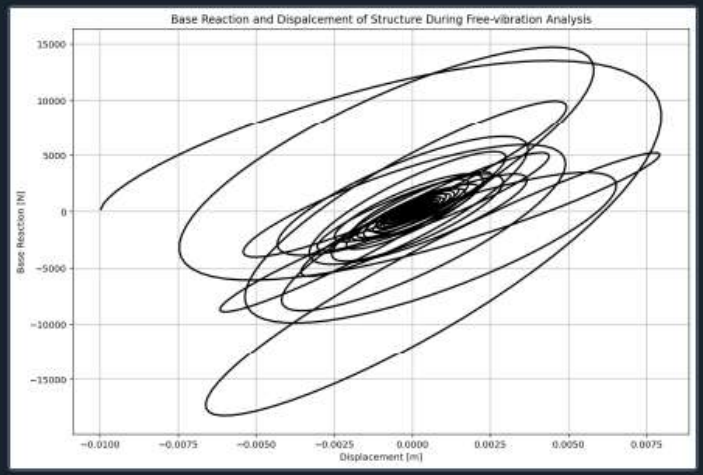
...GY_DISSIPATION_CAPACITY_INDEX\23_SOIL_STRUCTURE_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TURE_MDOF_8_ANA\P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

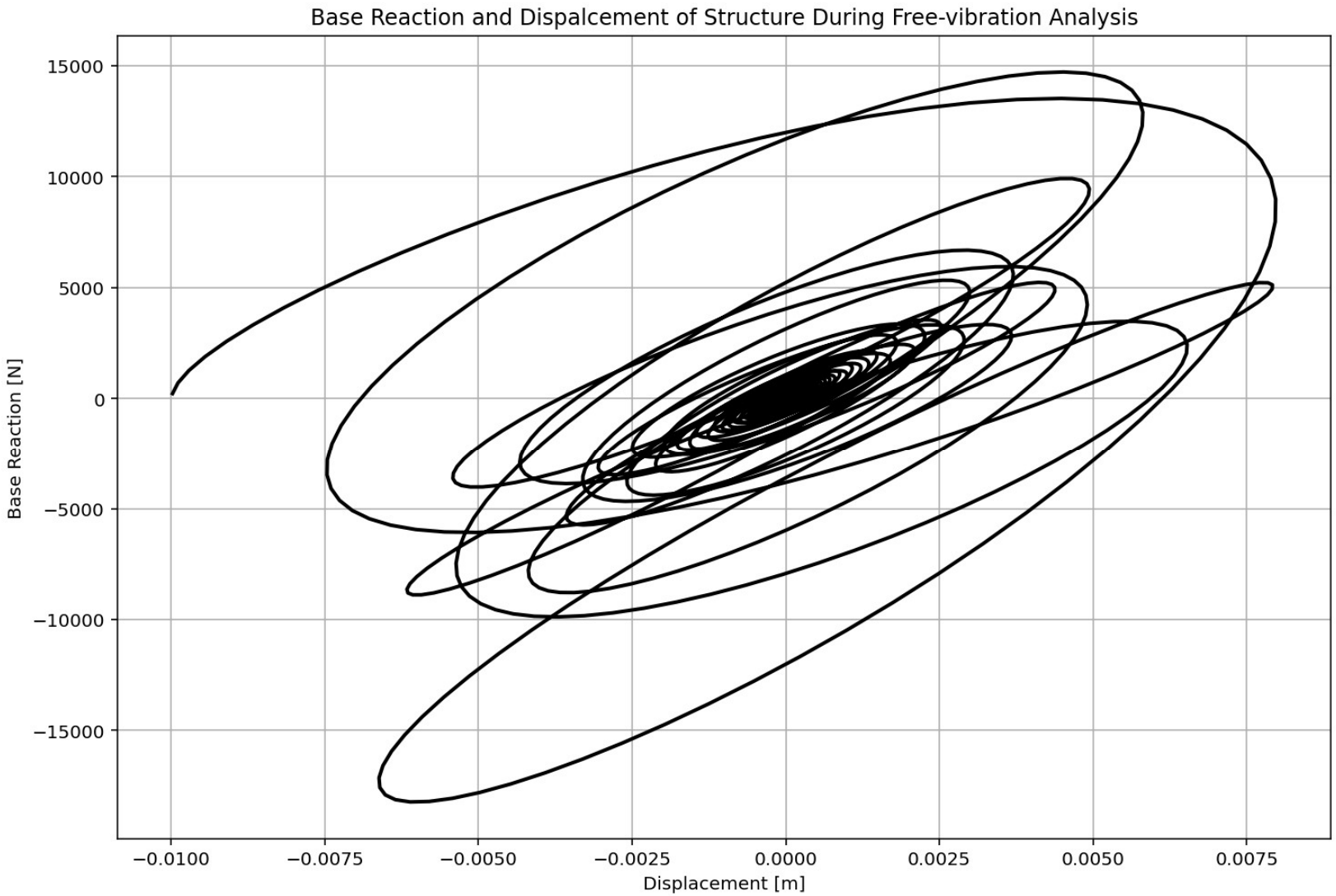
```
606     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
607         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
608         u0 = -0.010                         # [m] Initial displacement
609         v0 = 0.000015                       # [m/s] Initial velocity
610         a0 = 0.000065                       # [m/s^2] Initial acceleration
611         IU = True                           # Free Vibration with Initial Displacement
612         IV = True                           # Free Vibration with Initial Velocity
613         IA = True                           # Free Vibration with Initial Acceleration
614         duration = 20.0                     # [s] Analysis duration
615         dt = 0.001                          # [s] Time step
616         DR = 0.03                           # Damping Ratio
617
618         if IU == True:
619             # Define initial displacment
620             ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: https://openseespy
621         if IV == True:
622             # Define initial velocity
623             ops.setNodeVel(center_node, 1, v0, '-commit')  # INFO LINK: https://openseespy
624         if IA == True:
625             # Define initial  acceleration
626             ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: https://openseesp
627
628         ops.constraints('Plain')
629         ops.numberer('Plain')
630         ops.system('BandGeneral')
631         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://opens
632         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS - INFO LINK: http
633         alpha=0.5; beta=0.25;
634         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthed
635         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
636         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.read
637         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthedocs.io/en/late
638         ops.analysis('Transient') # INFO LINK: https://openseespydoc.readthedocs.io/en/lat
639
```

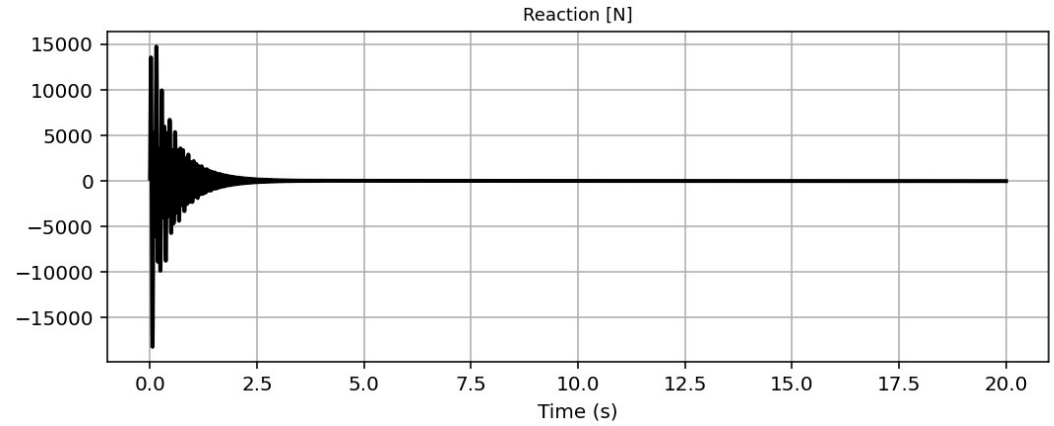
30 %

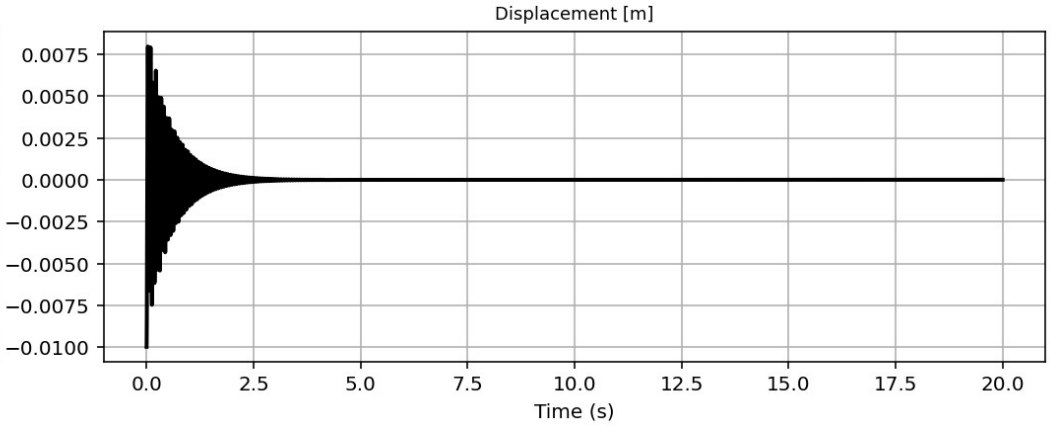


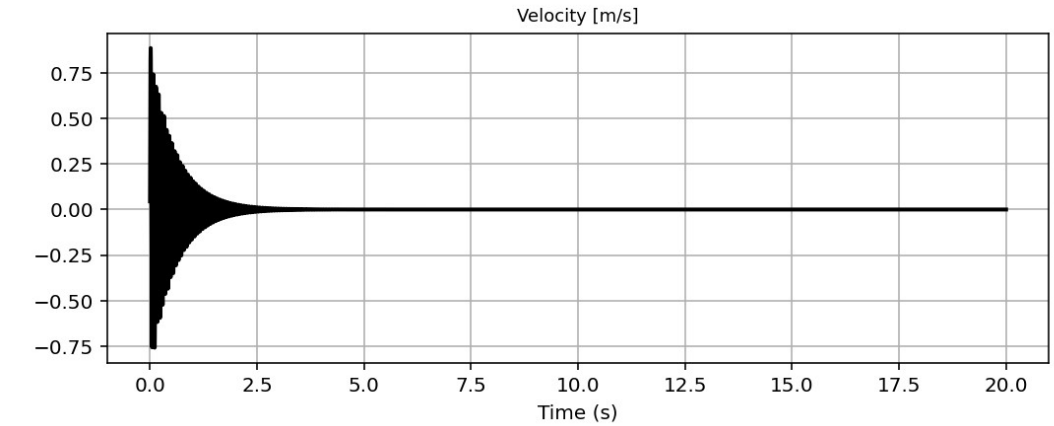
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

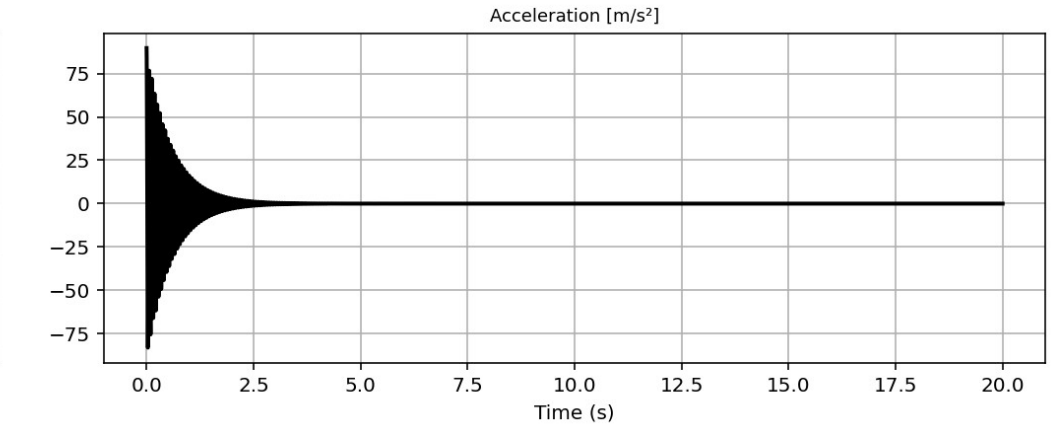
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 230, Col 55 UTF-8 CRLF RW Mem 47%

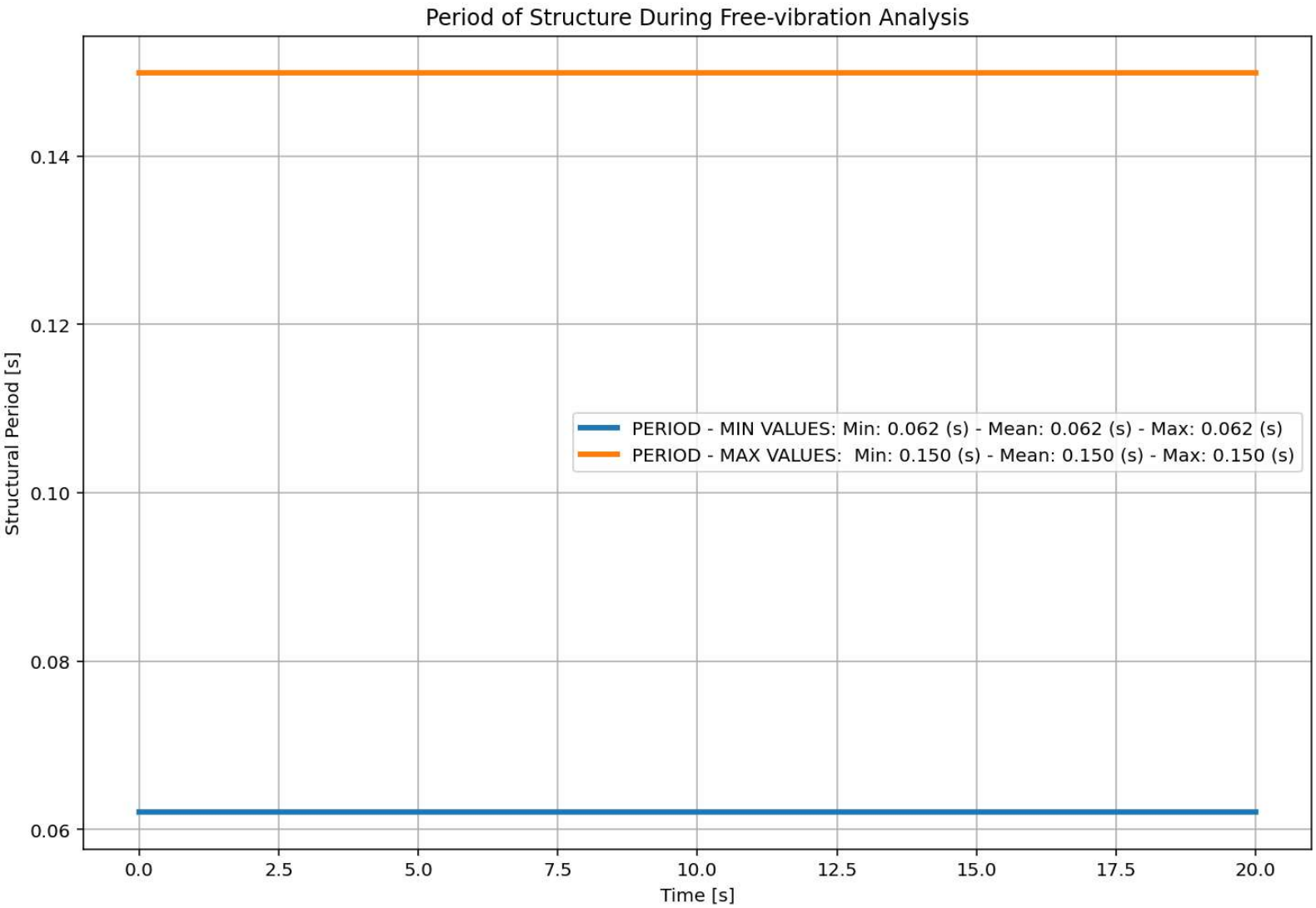

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
Displacement [m]

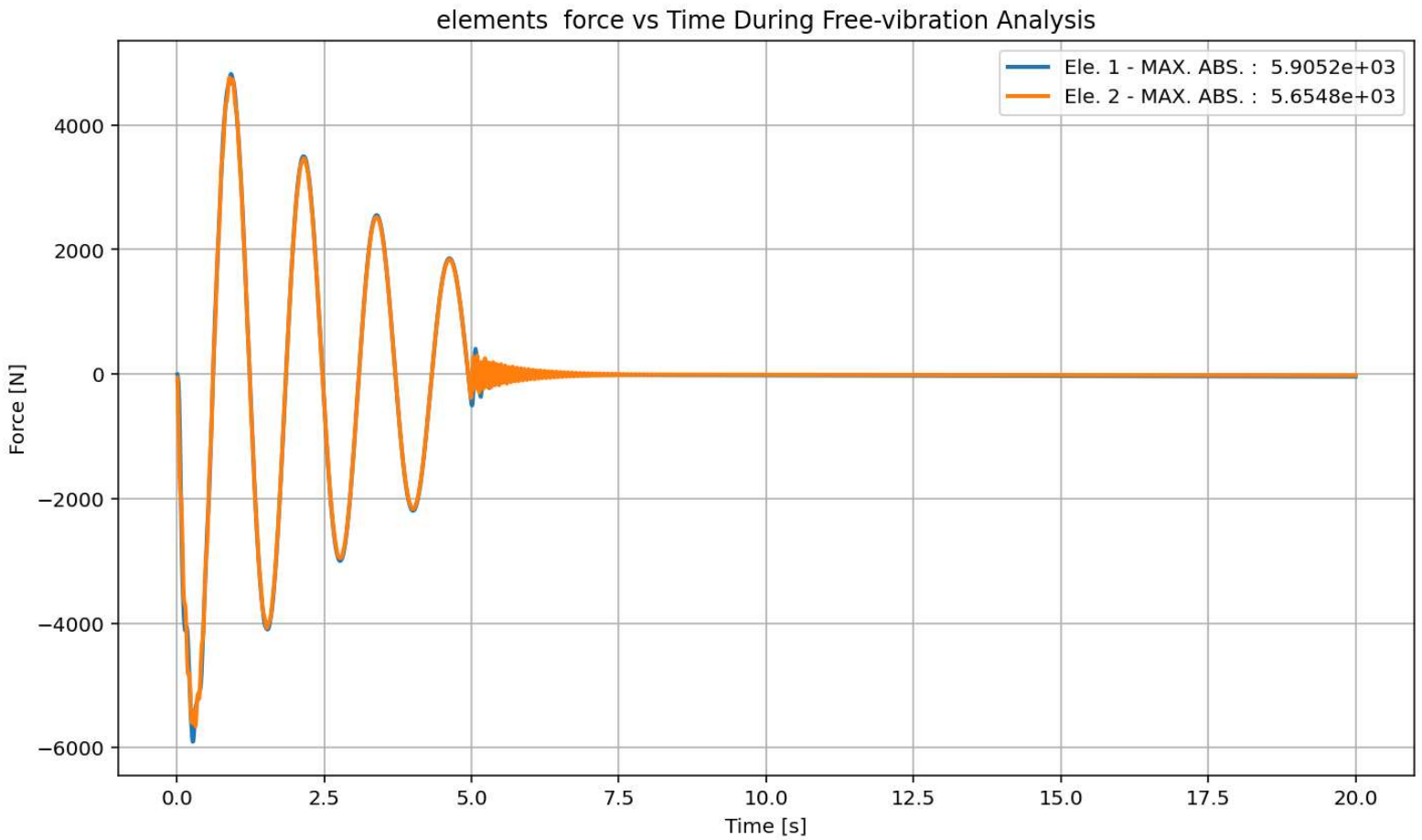

Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

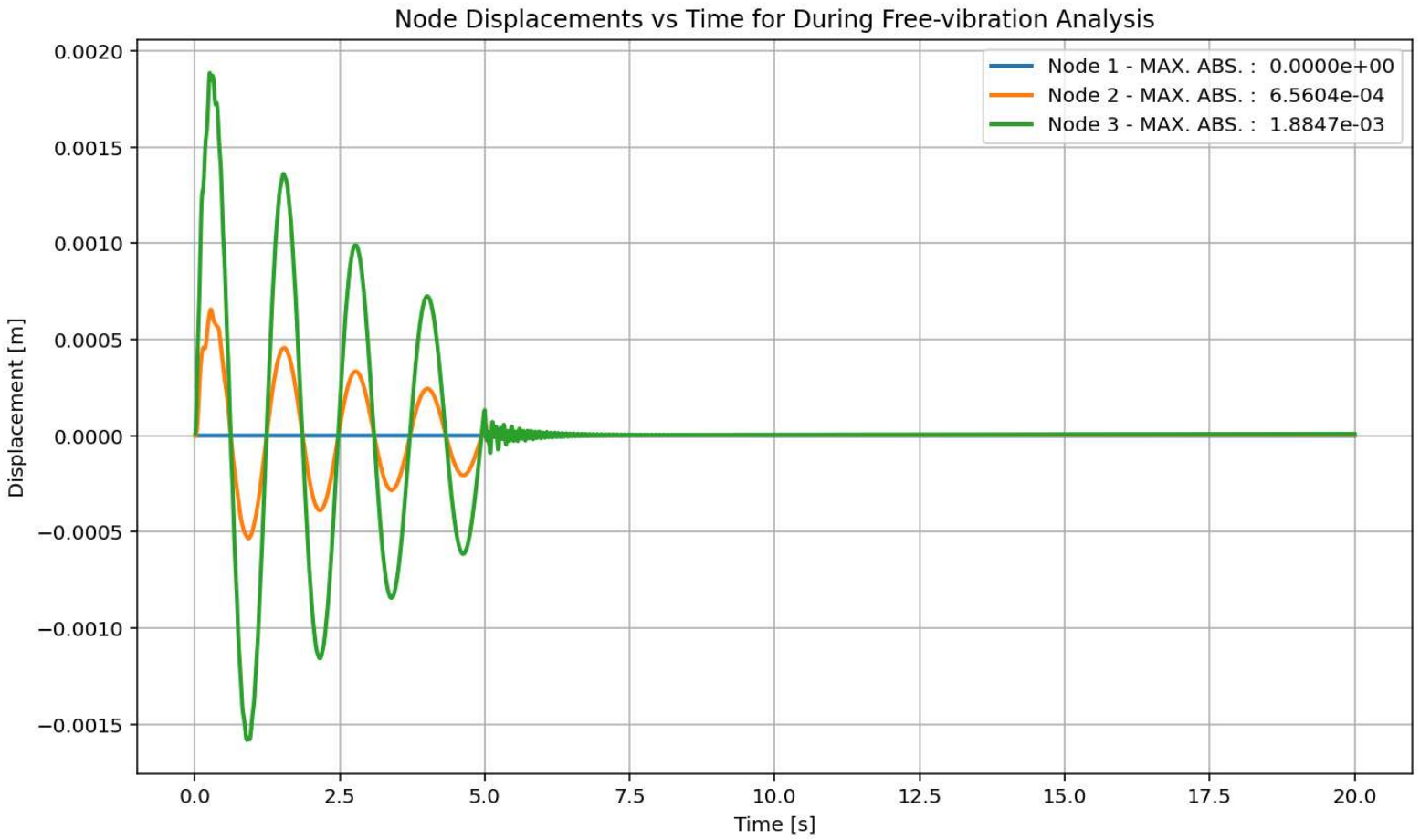

Displacement [m]
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

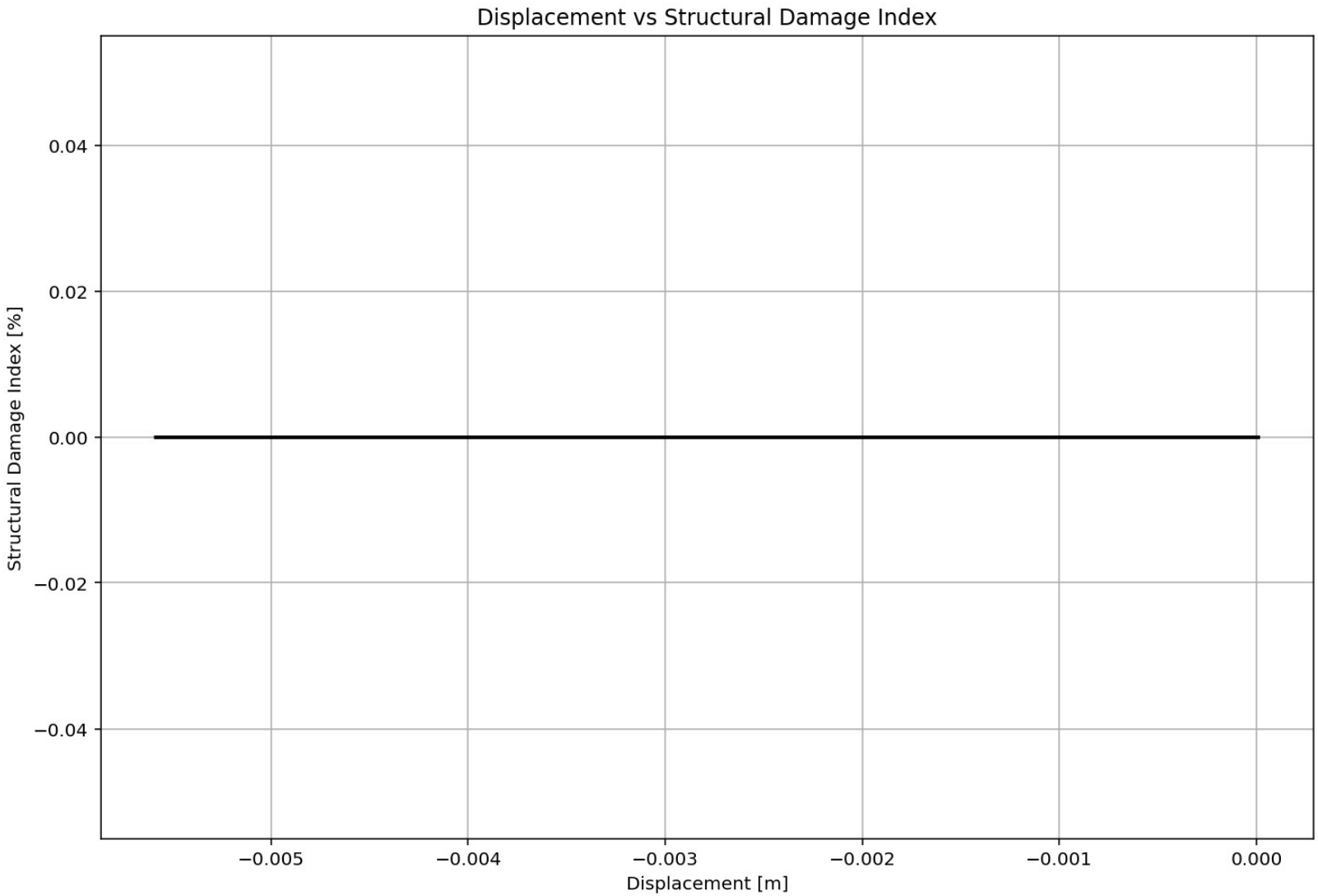

Velocity [m/s]
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

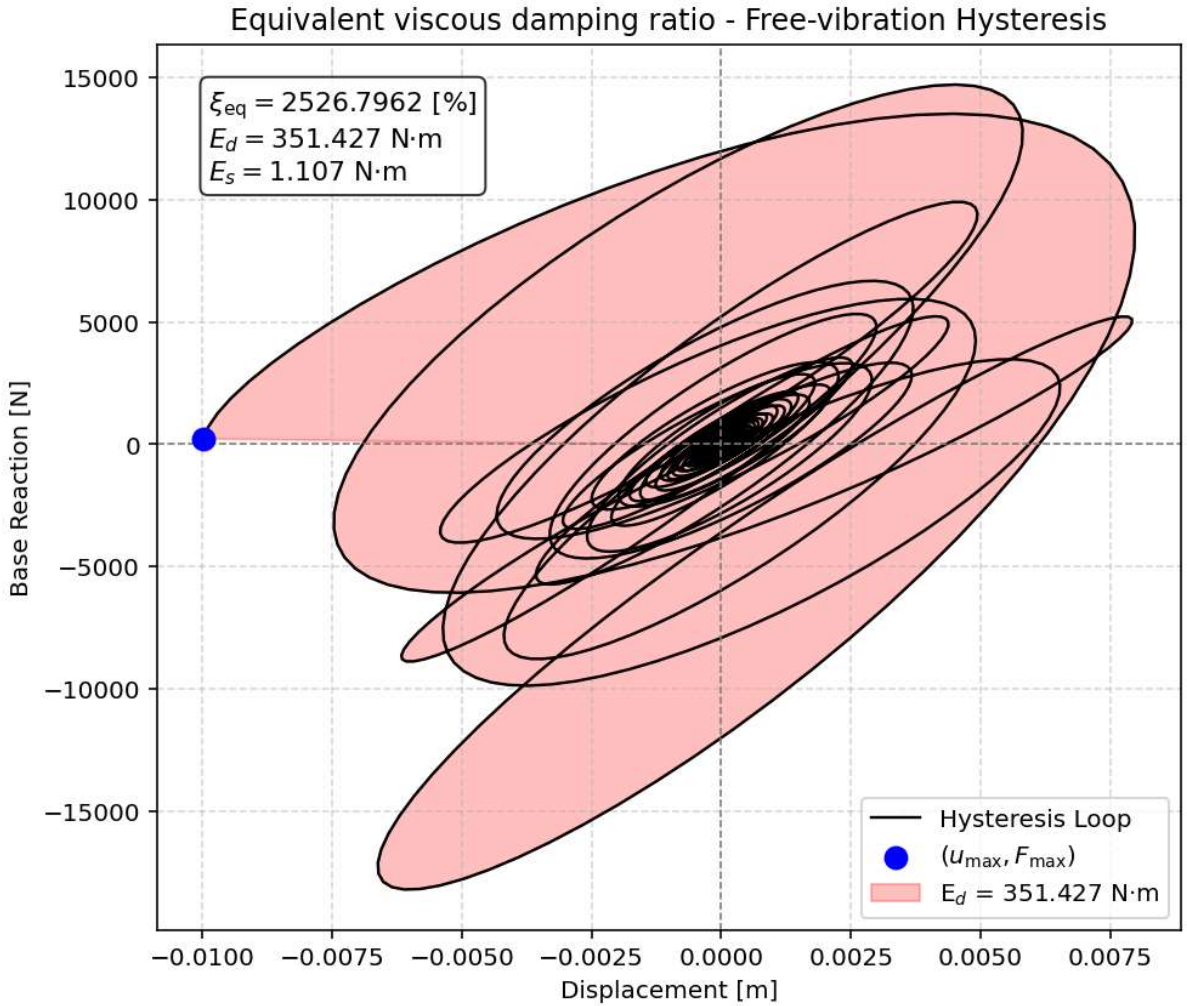

Acceleration [m/s²]
75
50
25
0
−25
−50
−75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Period of Structure During Free-vibration Analysis
Structural Period [s]
0.14
0.12
0.10
0.08
0.06
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.062 (s) - Max: 0.062 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 0.150 (s) - Max: 0.150 (s)


elements force vs Time During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 5.9052e+03
Ele. 2 - MAX. ABS. : 5.6548e+03
Force [N]
4000
2000
0
−2000
−4000
−6000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements vs Time for During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.5604e-04
Node 3 - MAX. ABS. : 1.8847e-03
Displacement [m]
0.0020
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.005
−0.004
−0.003
−0.002
−0.001
0.000
Displacement [m]


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 2526.7962$ [%]
$E_d = 351.427$ N·m
$E_s = 1.107$ N·m
Base Reaction [N]
Displacement [m]
15000
10000
5000
0
−5000
−10000
−15000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
Hysteresis Loop
(u_{max}, F_{max})
E_d = 351.427 N·m

SEISMIC ANALYSIS

Spyder (Python 3.12)

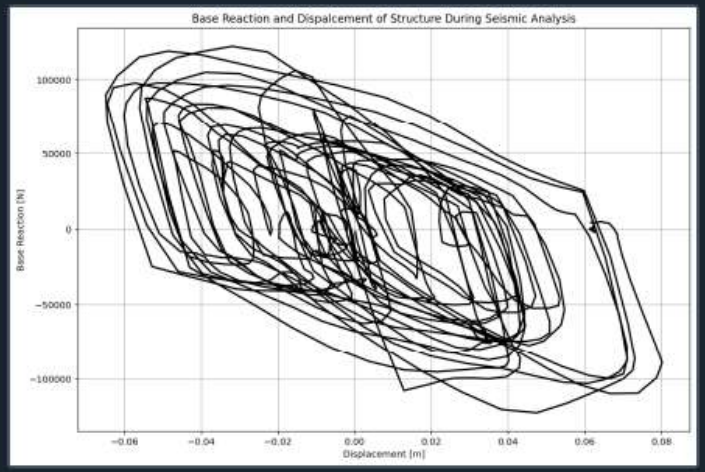
File Edit Search Source Run Debug Consoles Projects Tools View Help

...GY_DISSIPATION_CAPACITY_INDEX\23_SOIL_STRUCTURE_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TURE_MDOF_8_ANA\P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

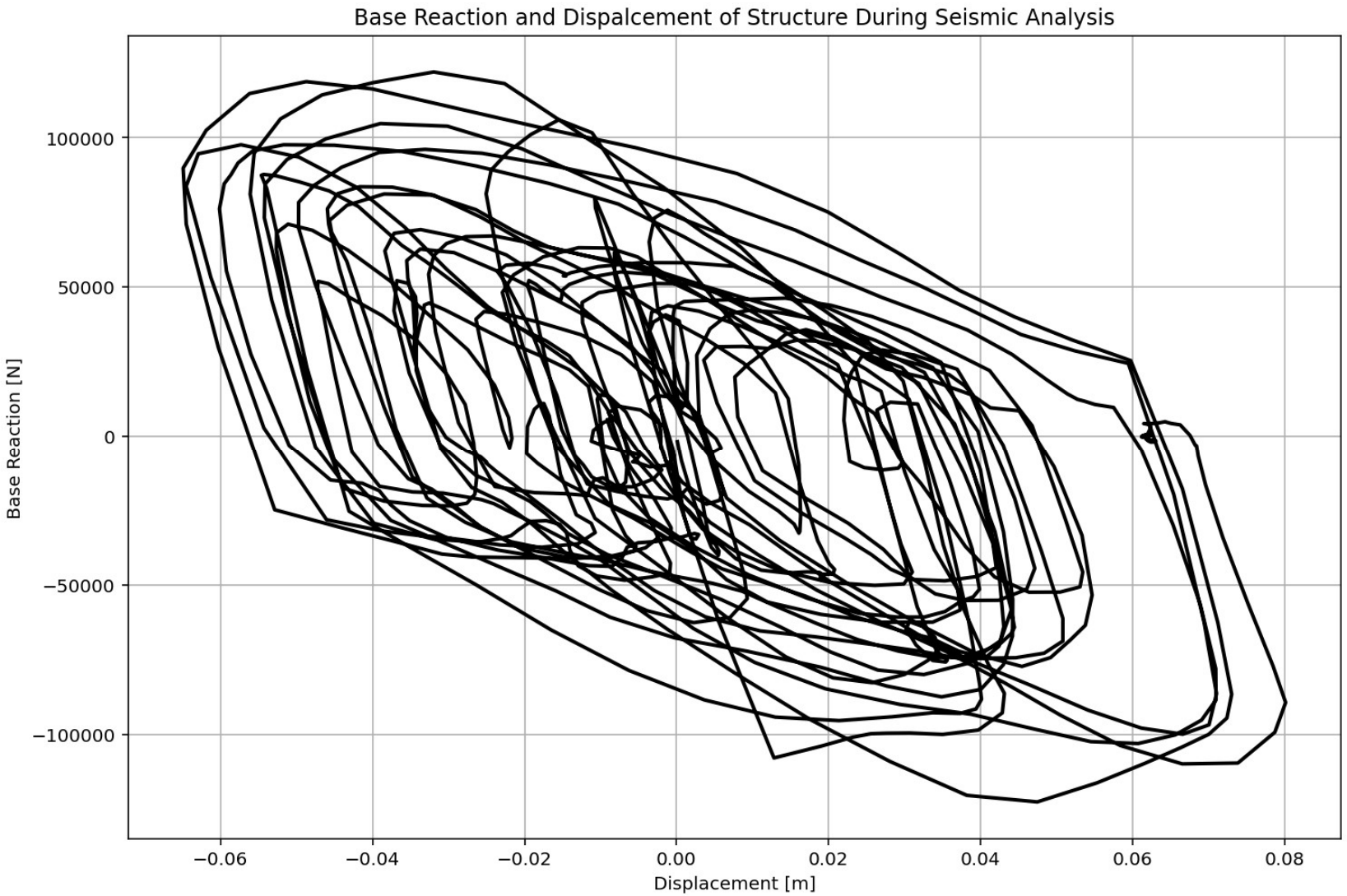
P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py

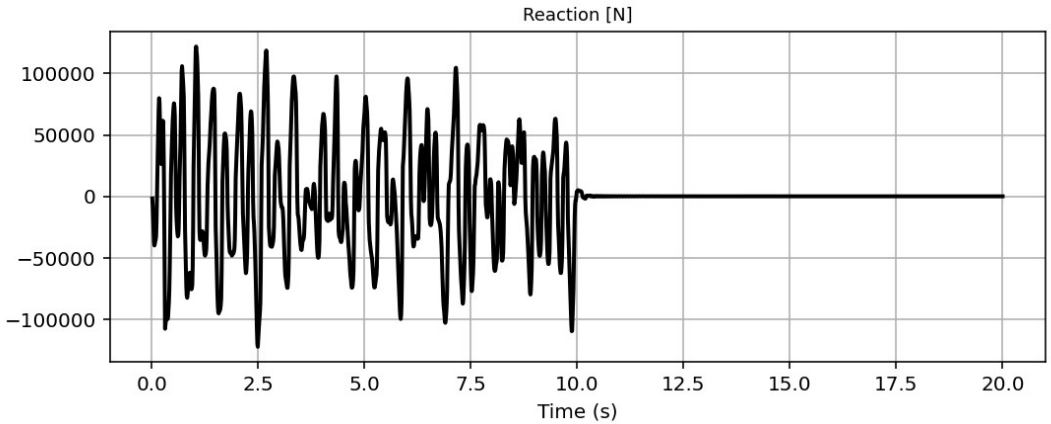
```
687     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
688         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
689         duration = 20.0                     # [s] Analysis duration
690         dt = 0.01                           # [s] Time step
691         GMfact = 9810                       # [mm/s²]standard acceleration of gravity or st
692         SSF_X = 0.10                        # Seismic Acceleration Scale Factor in X Direct
693         SSF_Y = 0.10                        # Seismic Acceleration Scale Factor in Y Direct
694         iv0_X = 0.0005                      # [mm/s] Initial velocity applied to the node
695         iv0_Y = 0.0005                      # [mm/s] Initial velocity applied to the node
696         st_iv0 = 0.0                        # [s] Initial velocity applied starting time
697         SEI = 'X'                           # Seismic Direction
698         DR = 0.03                           # Damping ratio
699
700         # Define time series for input motion (Acceleration time history)
701         if SEI == 'X':
702             SEISMIC_TAG_01 = 100
703             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
704             # Define load patterns
705             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
706             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
707         if SEI == 'Y':
708             SEISMIC_TAG_02 = 200
709             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
710             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
711         if SEI == 'XY':
712             SEISMIC_TAG_01 = 100
713             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
714             # Define load patterns
715             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
716             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
717             SEISMIC_TAG_02 = 200
718             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
719             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
720         print('Seismic Defined Done.')
```

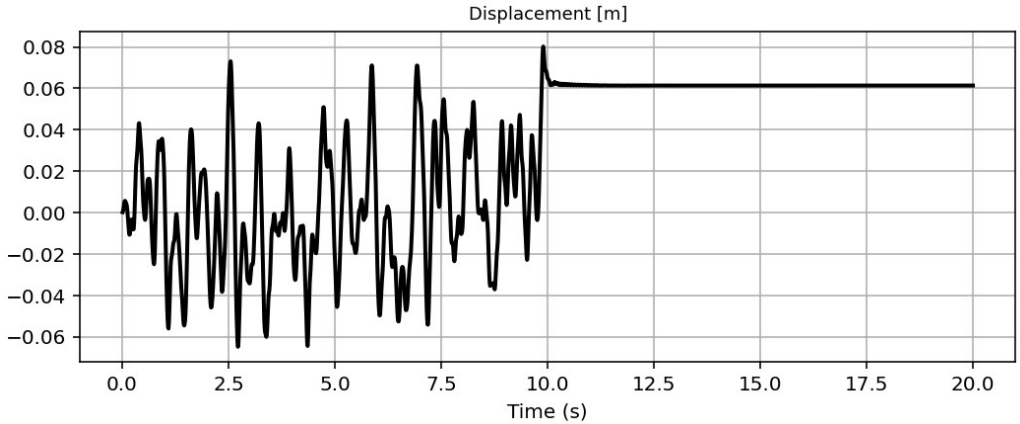


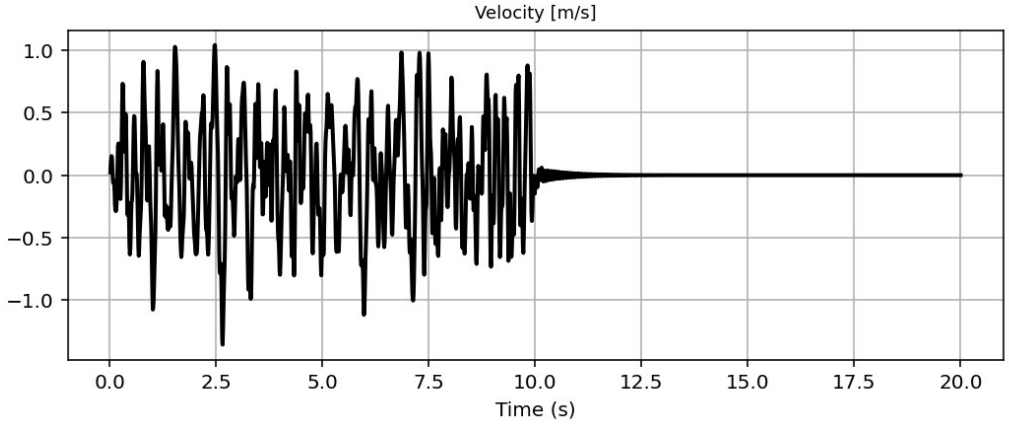
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

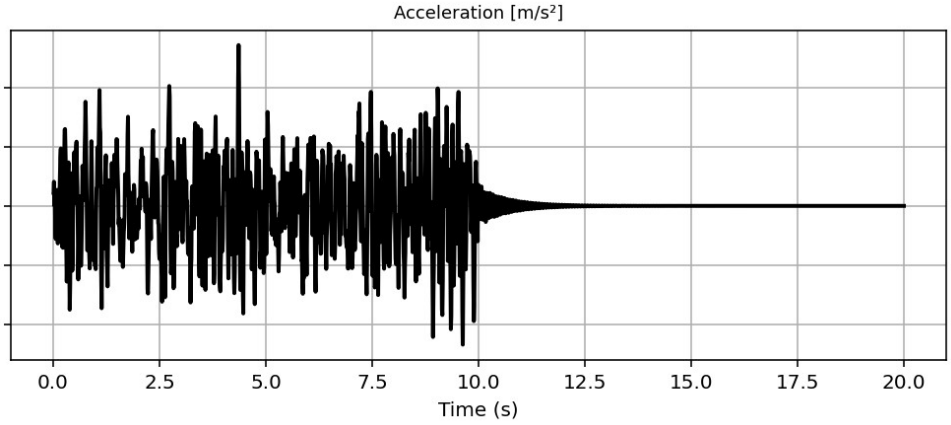
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 230, Col 55 UTF-8 CRLF RW Mem 47%

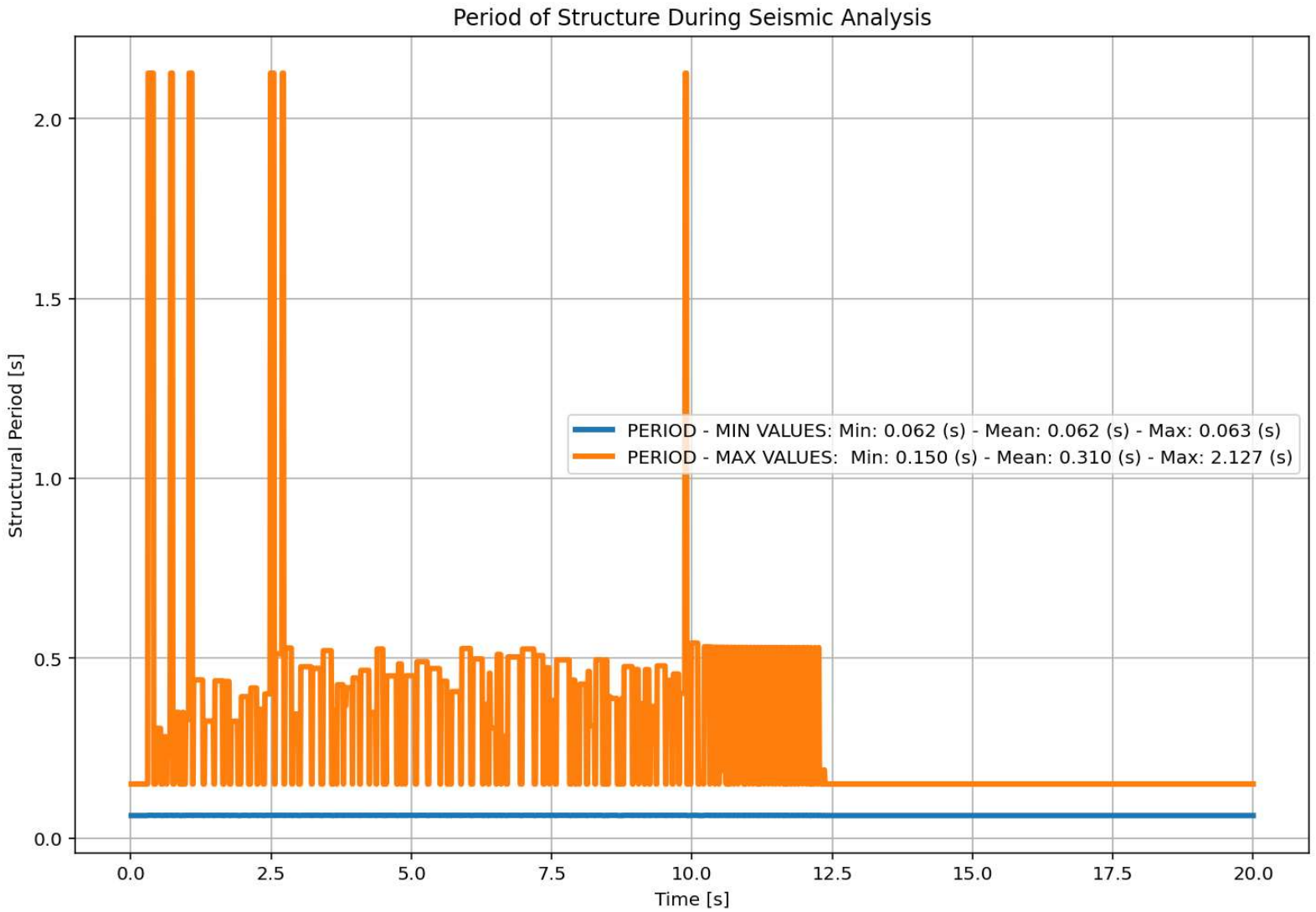

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
100000
50000
0
−50000
−100000
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Displacement [m]

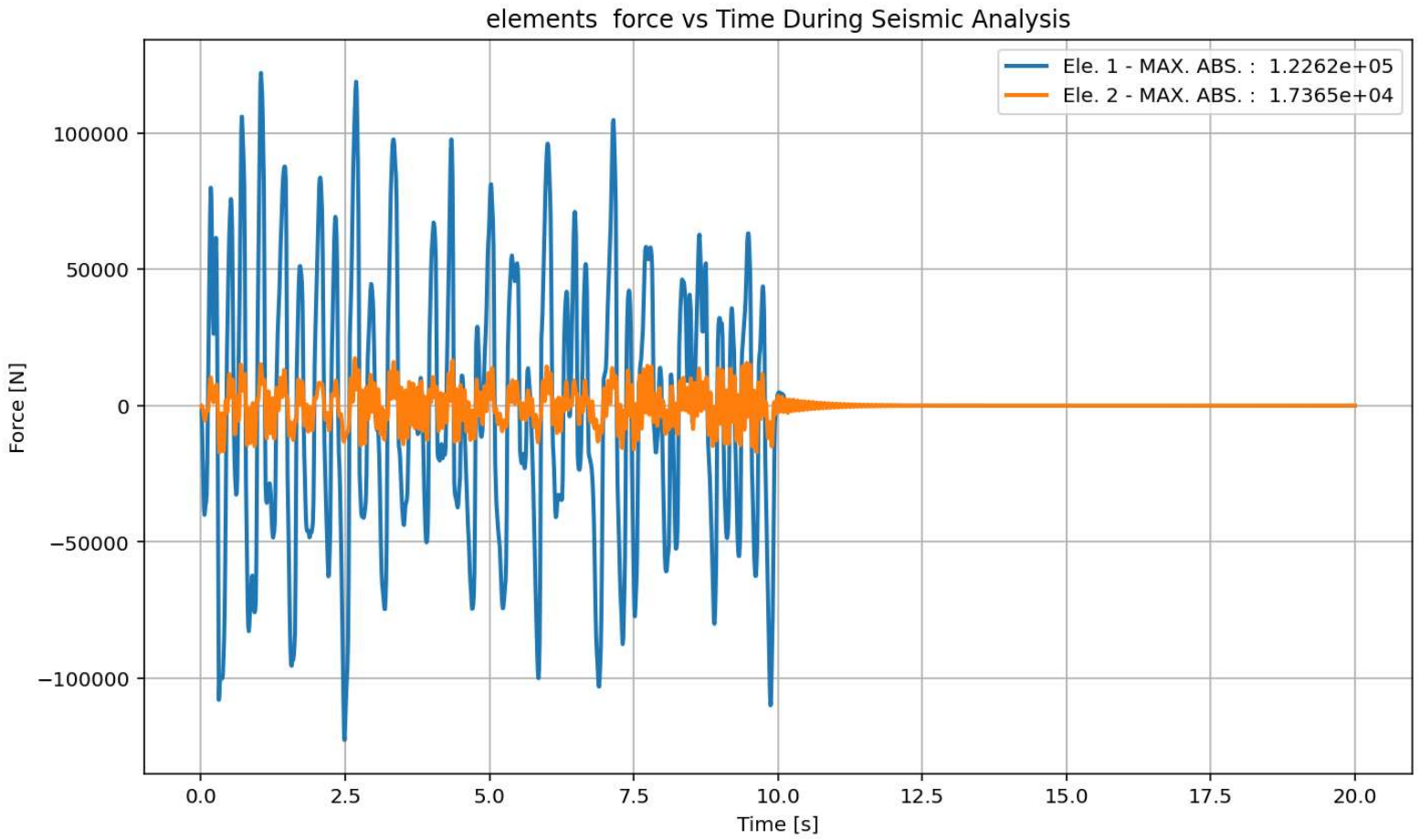

Reaction [N]
100000
50000
0
−50000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

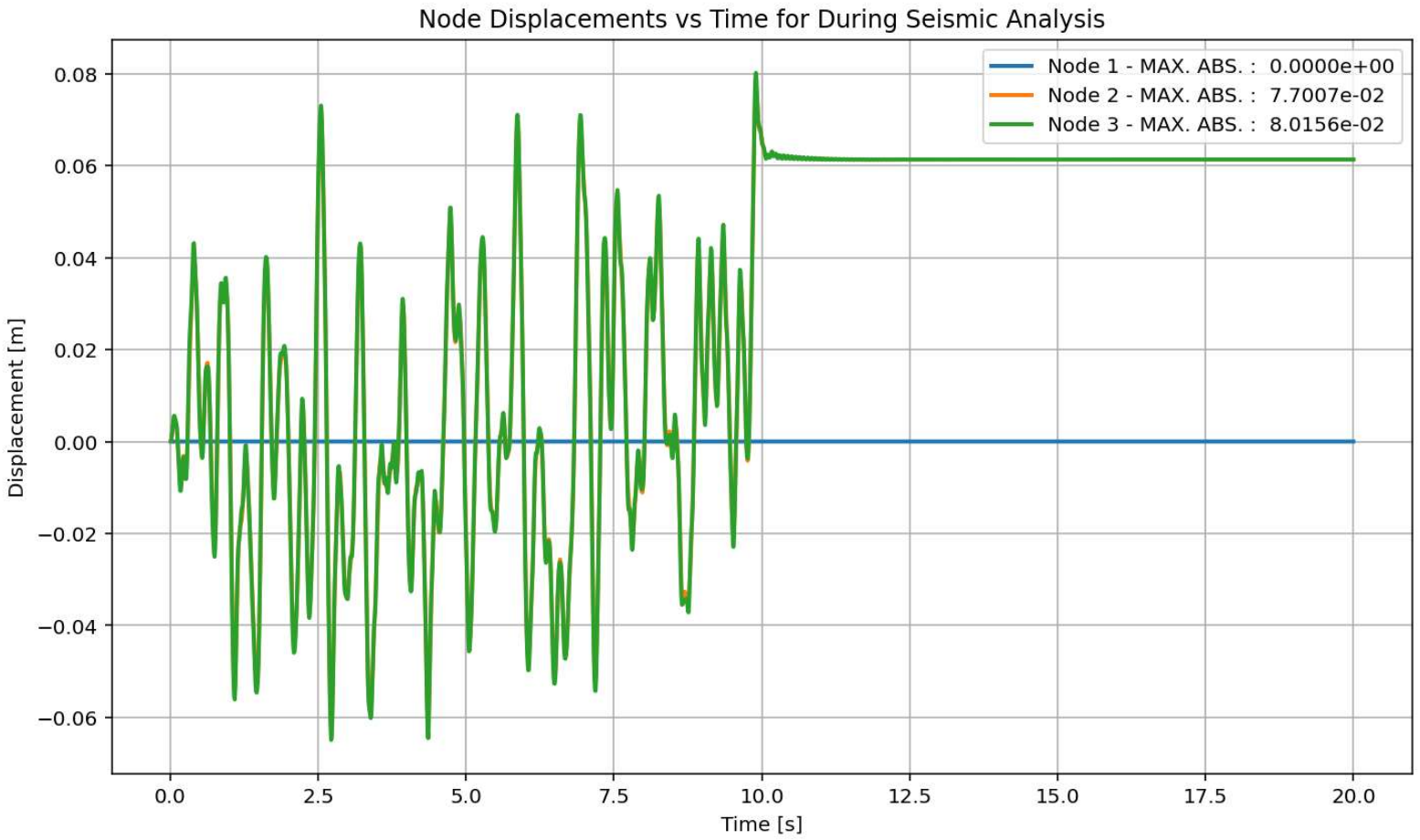

Displacement [m]
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

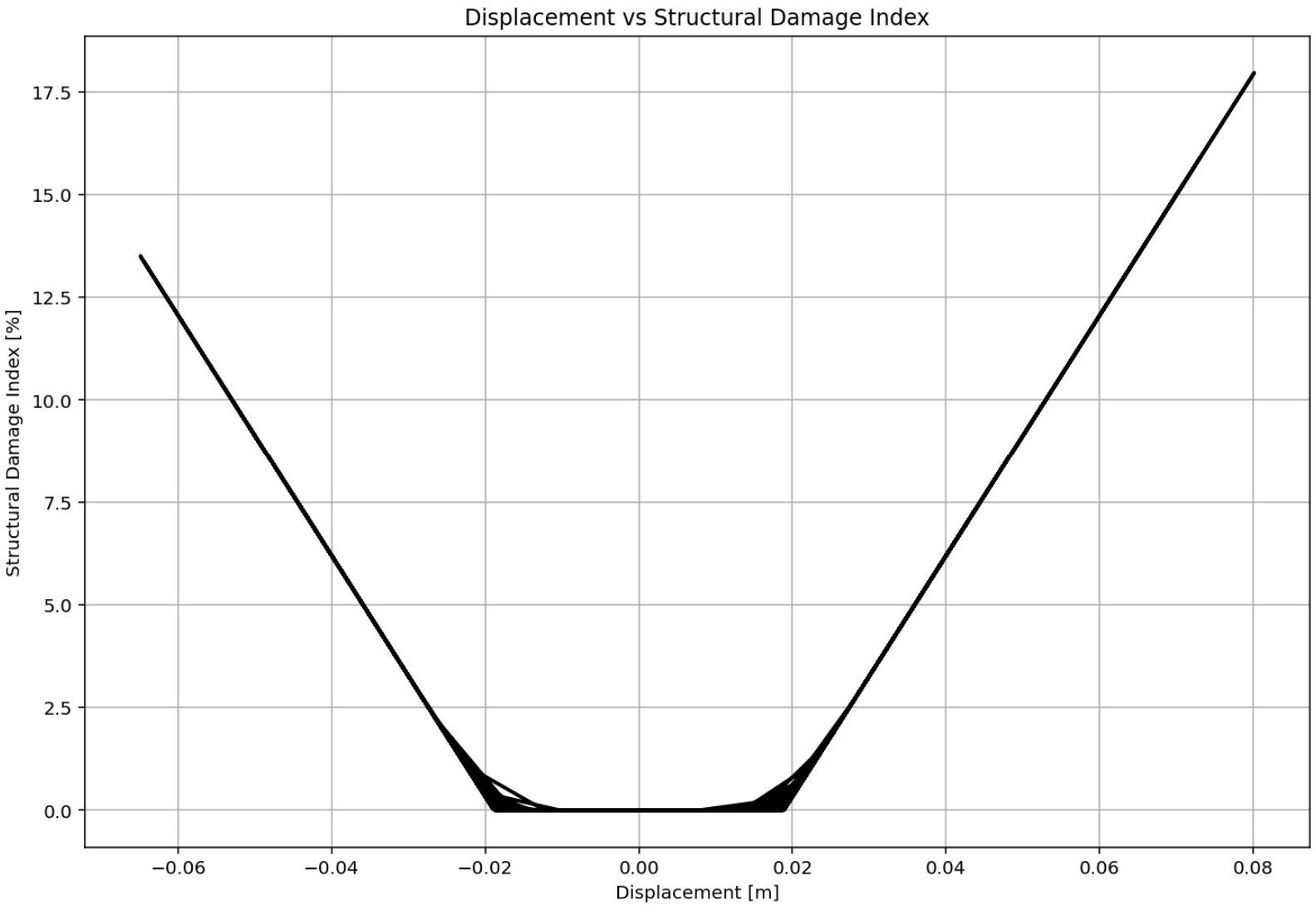

Velocity [m/s]
1.0
0.5
0.0
−0.5
−1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

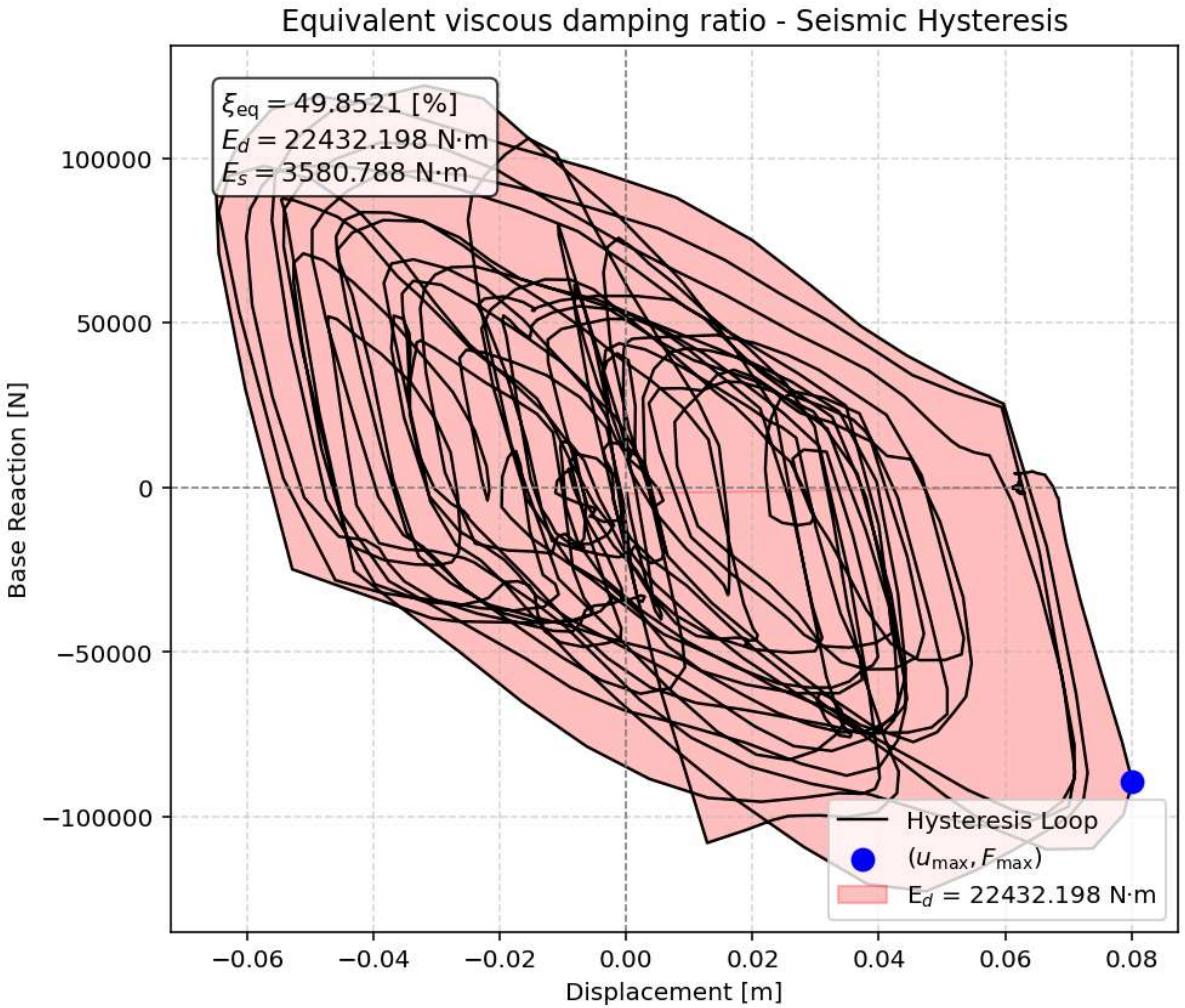

Acceleration [m/s²]
40
20
0
−20
−40
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

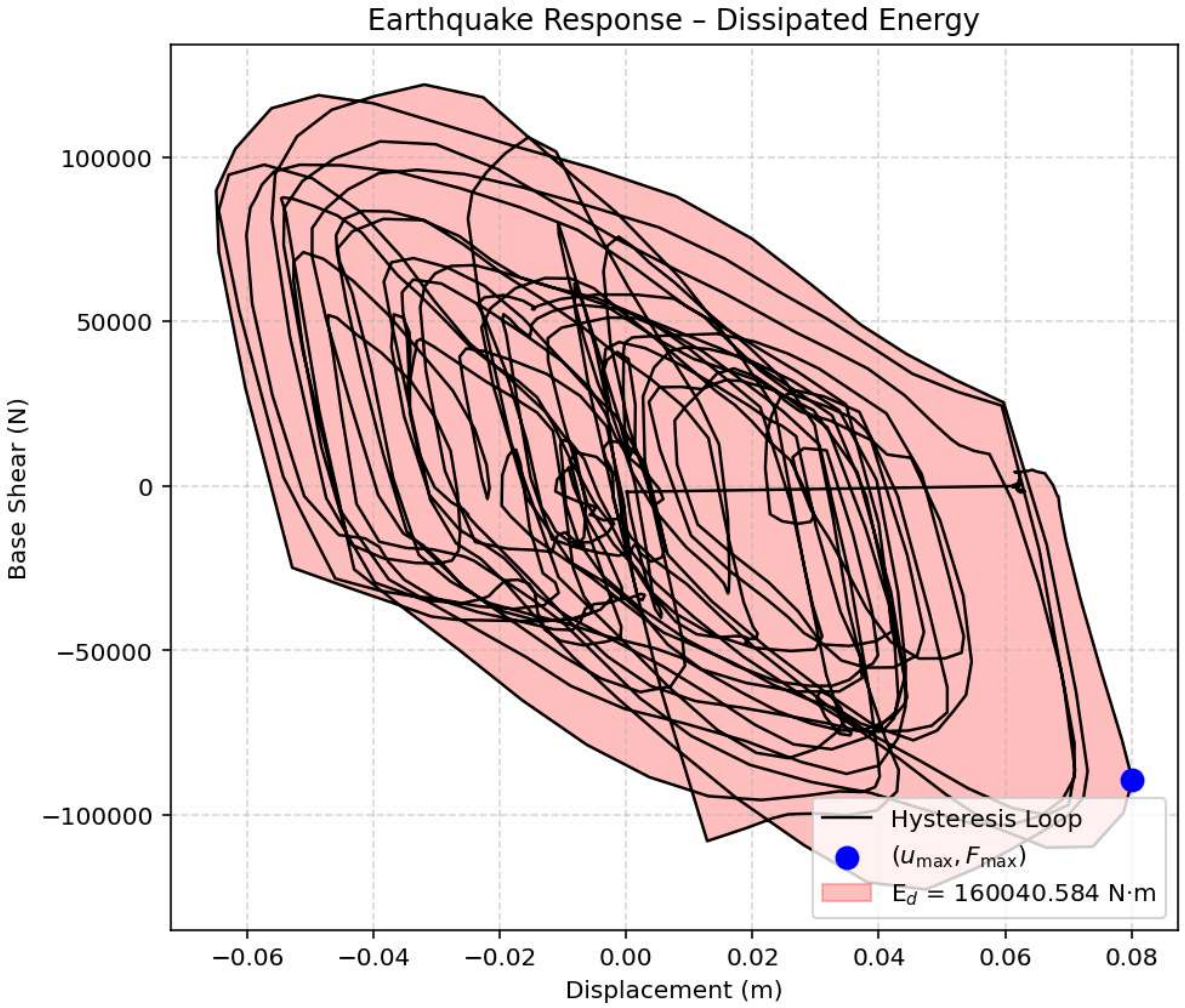

Period of Structure During Seismic Analysis
Structural Period [s]
Time [s]
0.0
0.5
1.0
1.5
2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
PERIOD - MIN VALUES: Min: 0.062 (s) - Mean: 0.062 (s) - Max: 0.063 (s)
PERIOD - MAX VALUES: Min: 0.150 (s) - Mean: 0.310 (s) - Max: 2.127 (s)

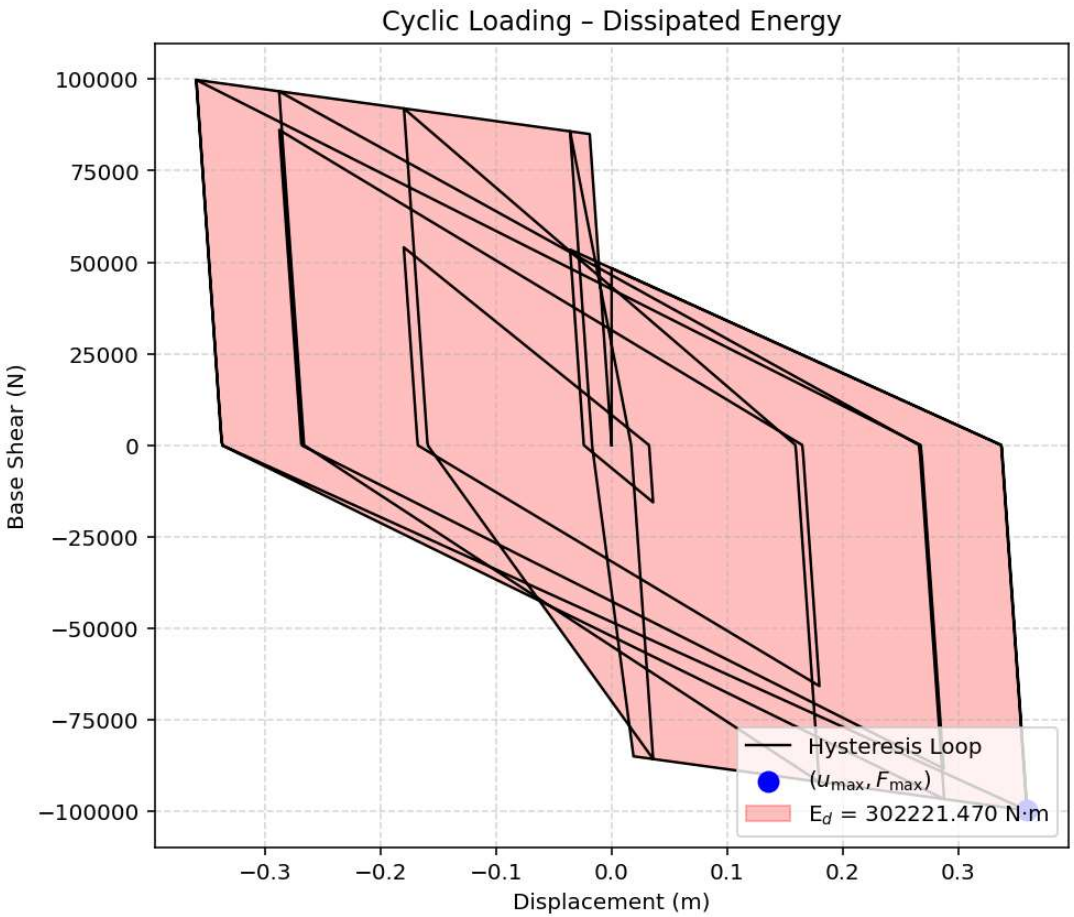

elements force vs Time During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.2262e+05
Ele. 2 - MAX. ABS. : 1.7365e+04
Force [N]
100000
50000
0
−50000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements vs Time for During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.7007e-02
Node 3 - MAX. ABS. : 8.0156e-02
Displacement [m]
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Displacement [m]

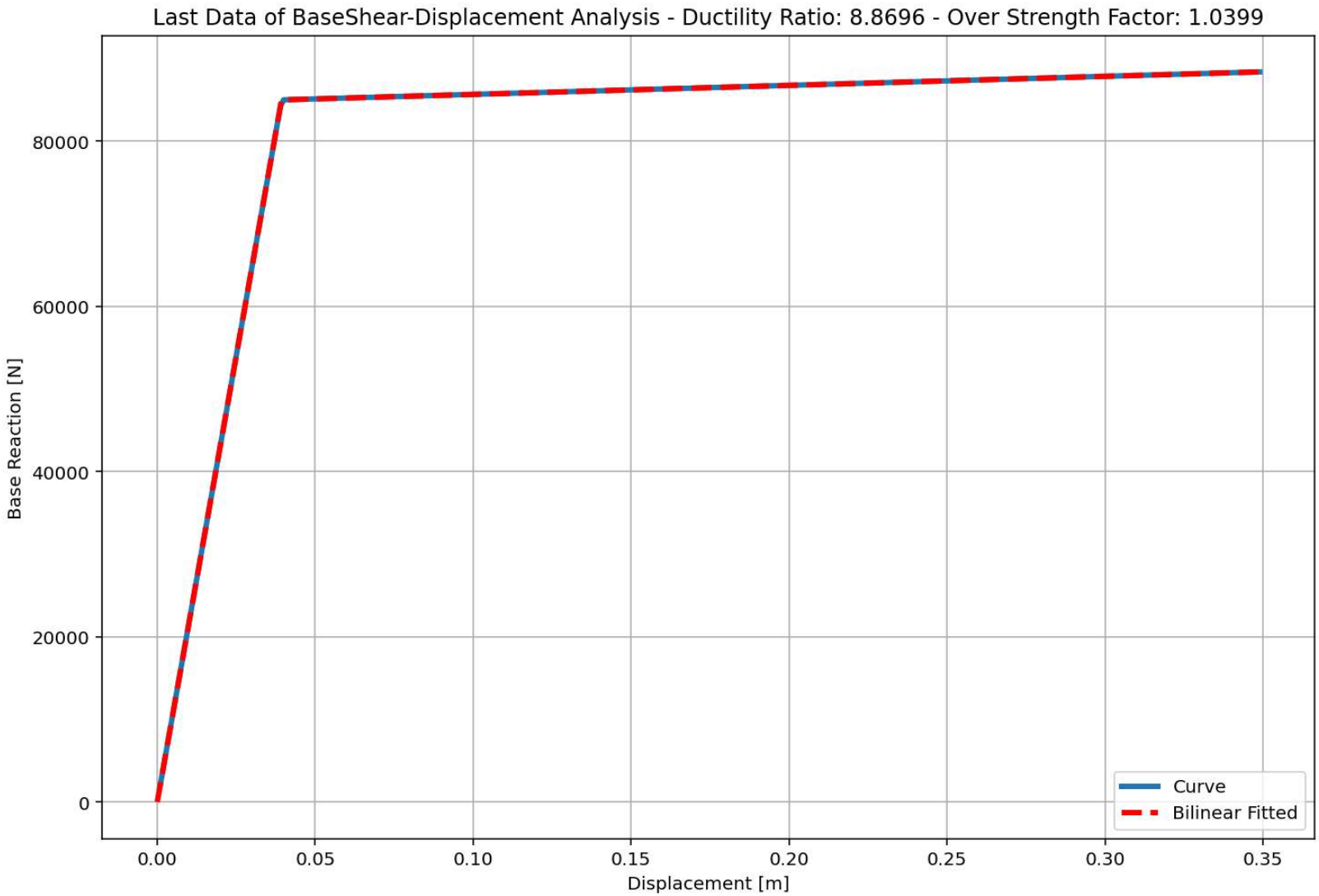

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 49.8521$ [%]
$E_d = 22432.198$ N·m
$E_s = 3580.788$ N·m
Base Reaction [N]
100000
50000
0
−50000
−100000
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 22432.198 N·m


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
50000
0
−50000
−100000
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Hysteresis Loop
(u_{max}, F_{max})
E_d = 160040.584 N·m

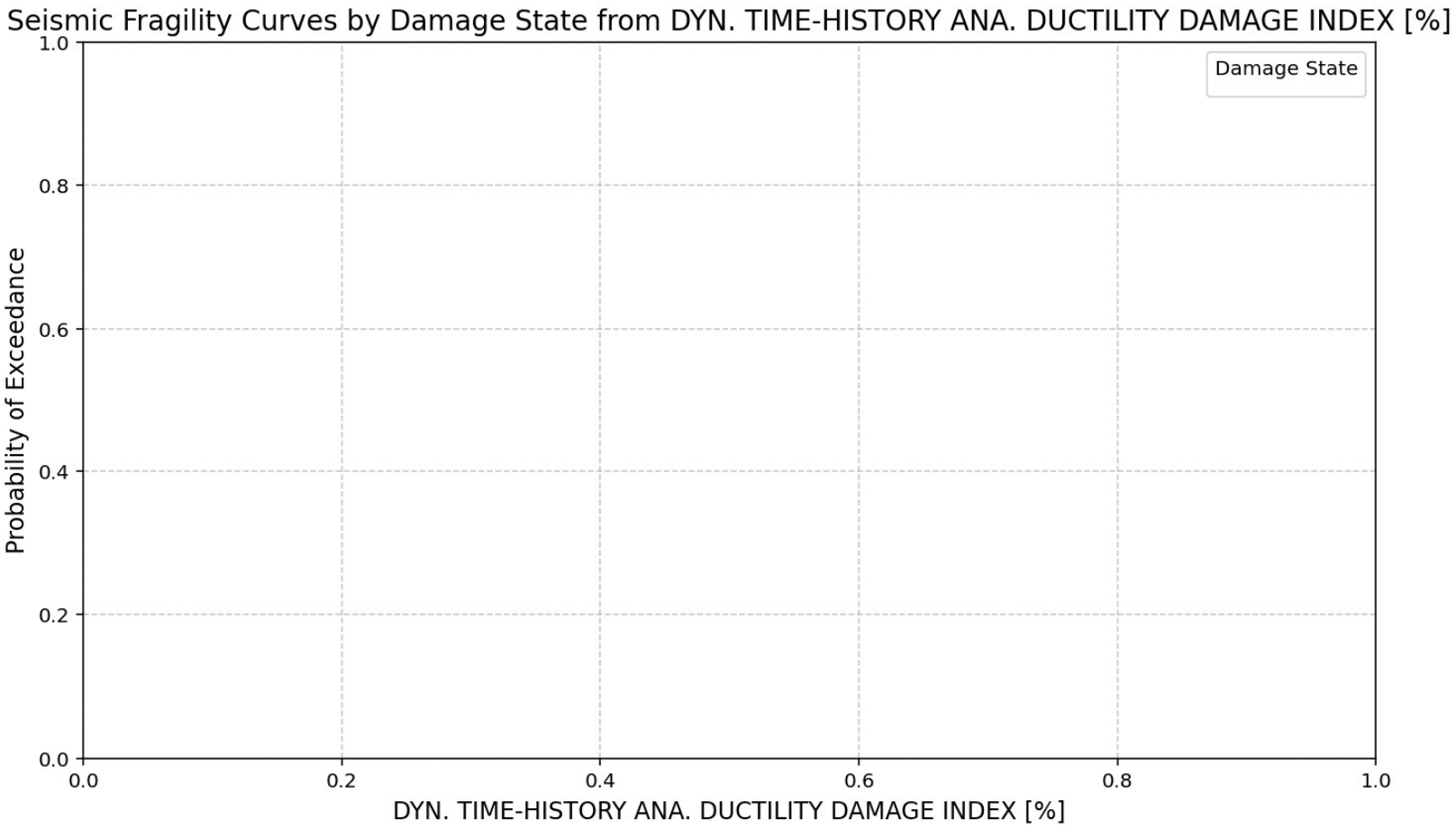

Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 302221.470 N·m

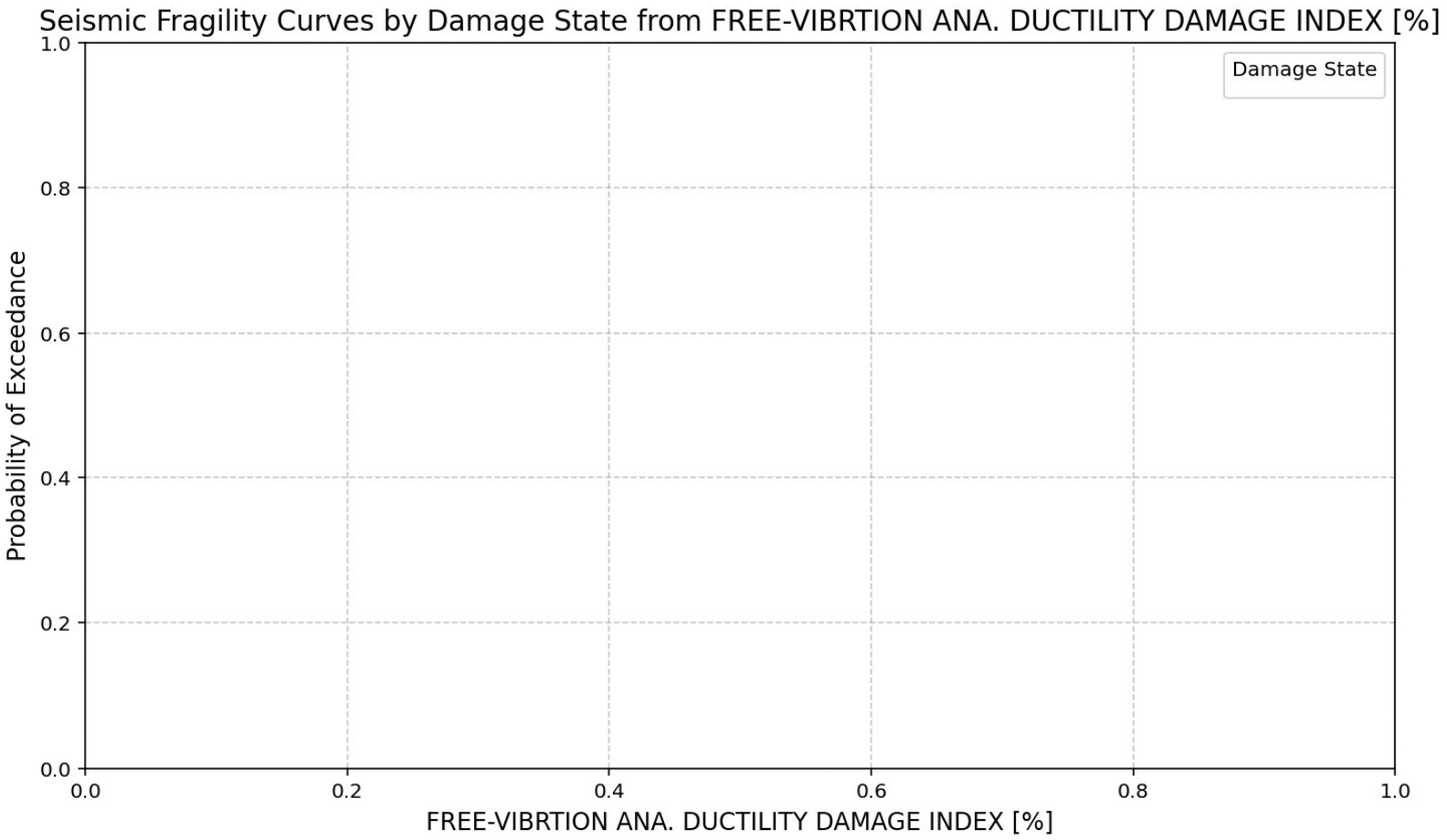
Dissipated Energy from Earthquake= 160040.58 N·m
Dissipated Energy from Cyclic Displacement= 302221.47 N·m

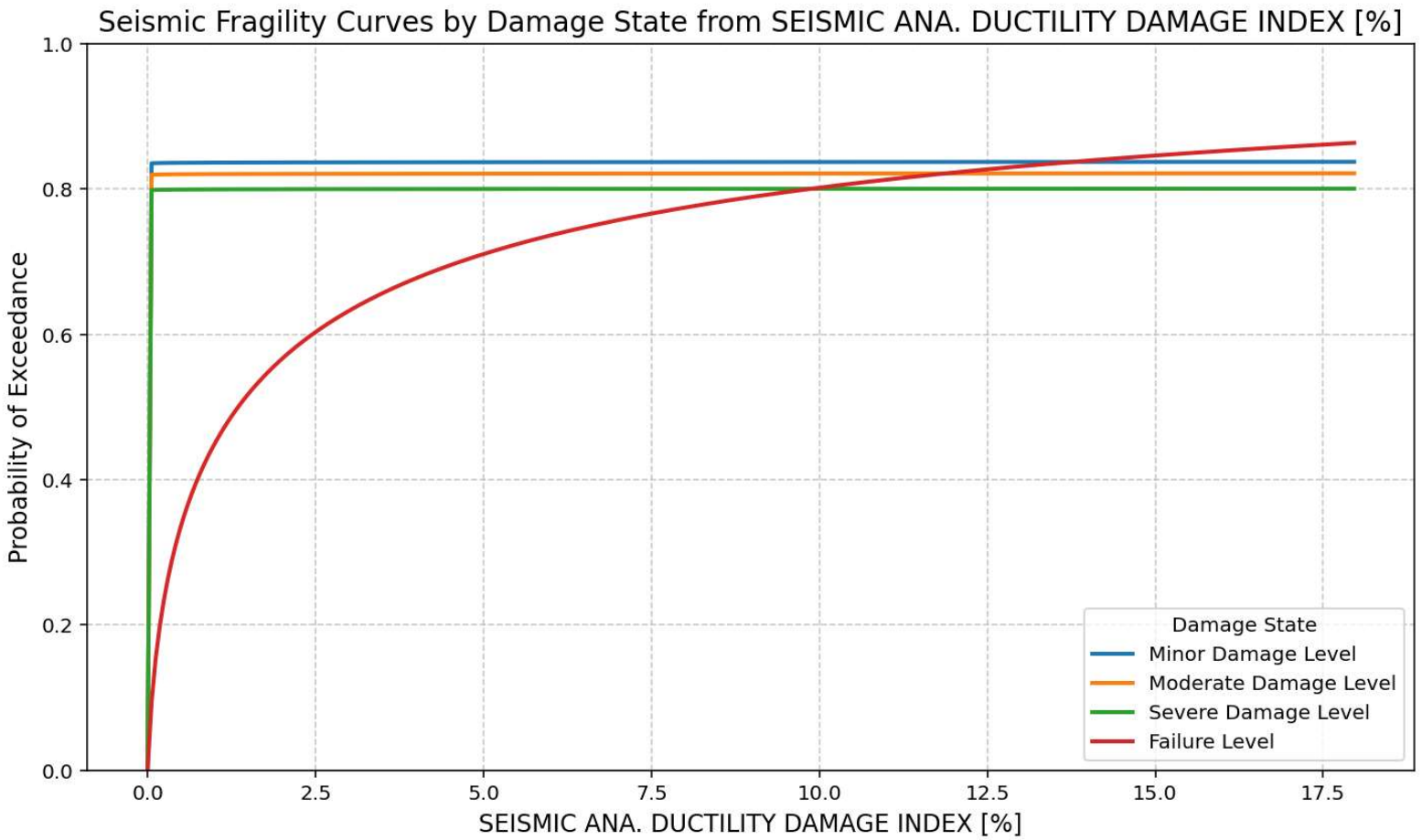
DISSIPATED ENERGY CAPACITY INDEX: 52.955 [%]
ZONE 6: SEVERE–LOW DAMAGE


Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 8.8696 - Over Strength Factor: 1.0399
Base Reaction [N]
80000
60000
40000
20000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement [m]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.88855309e-02 8.49848890e+04]
 [3.49000000e-01 9.93381699e+04]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    4500000.00
 Structure Plastic Stiffness :    284636.59
 Structure Tangent Stiffness :    43479.71
 Structure Ductility Ratio :      18.48
 Structure Over Strength Factor:  1.17
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     1.1745
Displacement Ductility Ratio (μ):   19.0621
Ductility Coefficient (Rμ):         19.0621
Structural Behavior Coefficient (R): 22.3886
Structural Ductility Damage Index in X Direction: 17.9620 (%)
```


Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level